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DESIGN AND MANUFACTURING OF A MODULAR UGV RESEARCH PLATFORM FOR CHALLENGING TERRAIN

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Abstract

Unmanned Ground Vehicles (UGVs) have become increasingly significant in both research and industrial applications due to their ability to operate in both hazardous environments and structured civilian settings. This project aims to develop a modular, teleoperated, and Robot Operating System 2 (ROS2) based platform to be used in future research project in the area of autonomous navigation. Furthermore, the platform will be designed to participate in Autonomous Robotic Experimental Systems (ARES) Makeathon. The process of manufacturing a UGV involves an interplay between mechanical structure, locomotion system, and control architecture. To facilitate the interplay of the subsystems, a comprehensive risk assessment was conducted as a preventative measure to mitigate risk. The risk assessment indicated that purchasing and modifying a pre-built platform would not be cost effective, therefore, it was decided to build a platform from scratch.

A prototype was designed and redesigned in accordance with the requirements of ARES and additional requirements from the stakeholders. The final design was manufactured using suitable materials to mitigate the impact of the operating environment. The platform uses a four wheel drive and suspension to control each wheel independently; this is not only effective when navigating rough environments, but also leaves more room for sensors in the chassis since the motors are mounted in the wheel rims.

The resulting platform weighs 49.2 kg, supports a minimum payload of 25 kg, operates for approximately 60 minutes, and successfully traverses slopes of $> 30°$***. Additionally, the platform incorporates modularity and adjustability regarding both structural configuration and sensor integration as well as teleoperation, implemented via ROS2. The outcome of this project has provided a solid UGV to compete in ARES Makeathon and facilitates future research within autonomous navigation and other research areas.***

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List of Abbreviation

UGV ..... Unmanned Ground Vehicle
ROS2 ..... Robot Operating System 2
ARES ..... Autonomous Robotic Experimental Systems
EDA ..... European Defence Agency
GPS ..... Global Positioning System
LiDAR ..... Light Detection and Ranging
AI ..... Artifical Intelligence
4WD ..... Four-wheel drive
PCB ..... Printed Circuit Board
PDB ..... Power Distribution Board
WBS ..... Work Breakdown Structure
CAN ..... Controller Area Network
RPM ..... Rounds Per Minute
ERPM ..... Electrical Rounds Per Minute
DC ..... Direct Current
BMS ..... Battery Management System
PLA ..... Polyactic Acid
CNC ..... Computer Numerical Control

1 Introduction

Unmanned Ground Vehicles UGVs have become increasingly significant in both research and industrial applications due to their ability to operate in environments that are hazardous or otherwise inaccessible to humans [1, 2, 3, 4]. Beyond hazardous applications, UGVs are also applied in more repetitive industries, such as agriculture [5]. For example, Quaglia et al. utilise UGVs for vineyard health monitoring, where the vehicle collects plant and soil samples and applies fertiliser autonomously. Similar principles are implemented in consumer-grade robotic systems, such as robotic lawn mowers, which rely on technologies such as autonomous navigation and object detection to perform their task [6, 7]. Given the expanding range of UGV applications, there is an increasing demand for platforms that are lightweight, capable, and adaptable. To address this need, this project focuses on the development of a UGV platform intended for research, defense, search-and-rescue operations, with particular emphasis on teleoperated use [4]. In such contexts, standardised interfaces and integration for sensors and computing units are crucial to ensure system flexibility, ease of maintenance, and future autonomous architectures.

A long-term objective of this platform is to serve as the foundation of a broader research initiative and to participate in the ARES Makeathon 2027. The European Defence Innovation Days ARES Makeathon is a competition organised by the European Defence Agency to promote EU-based startups developing Artifical Intelligence (AI) solutions for military decision-making [8]. In part, the design requirements have been derived from the official competition guidelines, with the primary requirements focusing on weight and dimensional constraints.

To ensure a high-quality final platform, the project follows an iterative prototyping approach along with a preliminary market survey. Initial prototypes are 3D printed to evaluate structural design aspects, such as suspension geometry and component fit. Based on the results of these tests, multiple design iterations are carried out before finalising the configuration. The final platform is manufactured using robust and lightweight materials such as aluminium and carbon fibre, which replace the temporary polymer-based components used in the prototypes. Off-the-shelf components were prioritised during development for reproducibility, availability, and spare parts.

The aim of this project is to develop a modular, teleoperated UGV platform that satisfies the defined project requirements listed in Table I and serves as the foundation for future integration of autonomous subsystems developed by a partner team. Within this scope, the platform is designed using modular principles and off-the-shelf components to facilitate reconfiguration, maintenance, and future expansion. Teleoperation using a handheld controller is investigated as the primary control method in this study.

2 Problem Formulation

The broader challenge described in Section 1 is developing a lightweight and modular UGV platform that is suitable for both the ARES Makeathon competition and future research. The construction of a comprehensive autonomous system with sensor integration is not in the scope of this project. Therefore, this study focuses specifically on the mechanical and teleoperational developments of the system. More precisely, this project addresses three core aspects:

1. Designing a lightweight and robust platform capable of meeting the requirements listed in Table I.
2. Developing modular mounting fastenings that support sensor integration.
3. Implementing a teleoperation framework using the ROS2 to enable reliable control.

The platform must not exceed a total mass of 50 kg or a combined dimensional sum (L + W + H) greater than 250 cm. Although a 50 kg weight limit is considered lightweight within the competition framework, the classification is relative. Quaglia et al. [5] constructed a UGV with a mass of 100 kg, whereas Rondelli et al. [9] showcased a platform with a mass between 630-920 kg (depending on additional equipment), both described as lightweight UGVs. Similarly, the Milrem THeMIS, with a total mass of 1630 kg, is also classified as lightweight [10]. This showcases the wide variation in how lightweight is defined within the UGV domain.

In addition to the ARES Makeathon competition constraints, the project scope is defined by stakeholder-specific requirements listed in Table I. These include the capability to traverse slopes of up to 30 degrees and to maintain wireless connectivity over a minimum distance of 100 meters without line-of-sight. Furthermore, the UGV is required to operate for at least 20 minutes under field conditions; however, this project targets an extended operational time of around 60 minutes. These constraints reflect realistic deployment conditions, in which standardised sensor mounting, rugged terrain capability, and ease of field repair are crucial for operational success. Employing modular and replaceable components, such as standardised fasteners and fittings, facilitates rapid repair and reconfiguration in the field. These requirements define the scope and limitations of the project and guide subsequent design decisions.

TABLE I
SYSTEM REQUIREMENTS

ID	Requirement Description
R1	The platform shall weigh no more than 50 kg.
R2	The platform shall be able to carry a minimum payload of 25 kg.
R3	The total measure (L + W + H) of the platform shall not exceed 250 cm.
R4	The platform shall be a four-wheel drive (Four-wheel drive (4WD)).
R5	The platform shall have suspension of any kind.
R6	The platform should be able to keep a max speed of 1 m/s.
R7	The platform shall be able to operate for at least 20 minutes.
R8	The platform must be able to traverse a 30-degree gradeability.
R9	The platform shall be able to move effectively in challenging terrain.
R10	The platform must include an emergency stop button

The defined scope and requirements represent the aspects of UGV development that this project can influence directly within the given resources and time frame.

RQ1: How can a UGV platform be designed to satisfy the requirements listed in Table I?

RQ2: How can modularity and adjustability be incorporated into the platform design with respect to structural configuration and sensor integration?

RQ3: How can the platform be teleoperated using the ROS2 framework?

To answer these questions, theoretical designs must be implemented and tested to evaluate whether the outcome of this project fulfils the expectations stated by the stakeholders.

3 Background

UGVs are robotic platforms designed to operate on land without an onboard human operator. Their control can range from higher levels of autonomy that include perception, planning, and control subsystems to direct teleoperation in which a human operator issues commands in real-time [1, 2, 3, 4]. Teleoperation is a control system in which, based on real-time sensor feedback, a human operator uses a wired or wireless connection to send motion and actuator commands. The vehicle may provide limited assistance, such as stabilising its movement or avoiding obstacles, but the operator remains in charge of all decisions [11, 12]. Initially, UGVs were created for military uses such as logistics, explosive ordnance disposal, and reconnaissance, where teleoperation was the primary control method for lowering personnel risk in dangerous situations [4]. Since many modern unmanned systems still function primarily at low levels of autonomy under direct human supervision, teleoperation continues to be a key mode of control in both military and civilian applications despite the rapid advancements in autonomy. At the same time, UGVs are being used increasingly in civilian fields such as disaster relief, infrastructure inspection, and agriculture [5, 9, 13].

3.1 Historical Development of UGVs

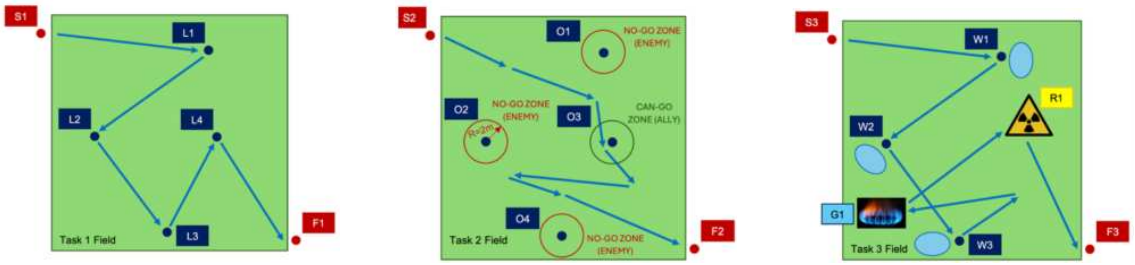
The origins of UGV development can be traced back to the early 20th century. Prior to that, Guglielmo Marconi showcased wireless telegraphy in 1896 [14] and Nikola Tesla secured a patent for a radio-controlled boat in 1898 [15], both depending on basic on/off signalling with restricted control functions. Furthermore, Leonardo Torres-Quevedo introduced the *Telekino* in 1903; a general architecture for transmitting coded instructions that could control multiple independent functions at a distance [16]. Following the Telekino, another milestone included DARPA's *Shakey* developed in the 1960s. This was recognised as the first mobile robot, which included integrated sensing, planning, and reasoning into a single architecture [17]. Developed at Stanford Research Institute International, Shakey combined television cameras, range sensors, and bump detectors with the Stanford Research Institute Problem Solver planner, which represented the world in symbolic terms and allowed high-level tasks to be decomposed into executable action sequences. This architecture enabled Shakey to navigate and avoid obstacles, demonstrating for the first time how perception could be tightly coupled with automated reasoning. Decades later, the Learning Applied to Ground Robots program (2004-2008) advanced vision-based navigation in unstructured terrain by supplying multiple research teams with identical small UGVs and evaluating progress through standardised field trials. Learning Applied to Ground Robots emphasised learning algorithms for long-range perception and adaptive obstacle avoidance, leading to major improvements in autonomous outdoor navigation [18, 19]. Together, these projects established fundamental principles of modern UGV autonomy: the integration of perception with symbolic reasoning, layered planning architecture, and learning-based adaptation to challenging terrain.

3.2 ARES Makeathon 2025

The ARES Makeathon is a European competition that promotes and focuses on the development and innovation of autonomous UGVs [20]. The 2025 competition was held at the Military University of Technology in Warsaw, Poland, and organized by the Centre for Mobile Robots in cooperation with the Ministry of National Defence. The event challenges university teams and startups to design and develop mobile robotic platforms capable of performing realistic defence-related missions. During the competition, participants completed three tasks:

1. **Last Mile Logistics** – Autonomous delivery of medical or logistical supplies while navigating checkpoints, avoiding obstacles, and utilising effective path-finding. This task is shown in Figure 1.1
2. **Reconnaissance Vehicle** – Autonomous patrol and identification of military objects and vehicles while avoiding restricted areas. This task is displayed in Figure 1.2
3. **CBRN sampling** – Operation of a robotic arm and sensing equipment to identify hazardous substances, perform pH measurements, detect gas leaks, and locate radioactive sources. This task is illustrated in Figure 1.3

These tasks collectively emphasise autonomy, perception, mobility, and environmental interaction, all of which are highly relevant to the development of modern UGV platforms.



1.1. Last Mile Logistics **1.2.** Reconnaissance Vehicle **1.3.** CBRN Sampling

Figure 1: ARES Makeathon 2025 tasks

3.3 Design Considerations of UGV Platforms

The design of a UGV involves an interplay between mechanical structure, locomotion system, and control architecture. Each design decision below influences mobility, energy efficiency, stability, and adaptability in different operating environments. Taking into account and addressing these interdependencies is a crucial design consideration when designing a UGV from scratch that is expected to perform reliably under the given requirements listed in section 1.

3.3.1 Modularity and Structural Optimisation

As the field developed, structural optimisation and modularity have progressively become more significant to UGV design considerations [21]. Modular platforms equipped with additional sensors such as cameras, Global Positioning System (GPS) units, and Light Detection and Ranging (LiDAR) systems are adaptable to a wide range of tasks without requiring extensive structural redesign. Such architectures have been demonstrated in different contexts, such as research and defence, where standardised mounting interfaces have enabled sensors and actuators to be swapped depending on the given task. Modularity increases flexibility and long-term maintainability, but it also adds weight and complexity, requiring trade-offs that need to be carefully managed under strict specifications. Additionally, structural optimisation regarding weight and stability has played an important role in enabling platforms to be lightweight without losing structural integrity [22]. For example, by introducing techniques such as finite element analysis, engineers can systematically remove non-load-bearing sections of a platform in order to optimise its overall structure and reduce overall weight without compromising the strength of the platform [23]. R1 in Table I requires the platform to weigh in at 50 kg, making such techniques critical when designing and manufacturing a platform from scratch that is within this constraint. By combining modular design with structural optimisation techniques, UGVs are able to achieve both adaptability and performance. This results in a robust platform while maintaining a lightweight platform in comparison to the overall market.

3.3.2 Mobility and Locomotion Trade-offs

Mobility directly affects where and how a platform can operate, making it a defining design aspect for a UGV. Wheeled platforms have the advantage of high speed and efficiency on flat surfaces, and even slightly challenging terrain, making them suitable and frequently used in logistical and agricultural contexts [24]. However, their performance is reduced in challenging terrain such as sand, loose soil, or steep slopes, where traction is typically more critical to overcome these conditions than speed. These limitations can be mitigated by choosing appropriate wheel types with the physical attributes to adapt to the conditions.

Wheeled platforms are adaptable and come in different configurations, such as a 4x4 or 6x6 drive. Tracked vehicles typically provide better stability by distributing their ground pressure over a larger contact area in comparison to wheeled vehicles, which in turn reduces sinkage on softer surfaces; both Wong [25] and Bekker [26] highlight this. On the other hand, weight, energy consumption, and mechanical deterioration are drawbacks of tracks. Tracked vehicles with passive or semi-active suspension offer better obstacle clearance, slope stability, and vibration damping than unsuspended models. In order to lessen these disadvantages, suspension systems are essential [27]. Hybrid mobility concepts, such as leg-wheel systems and adaptive wheel design, aim to combine the terrain versatility of tracks with the efficiency of wheels, thereby improving adaptability across various terrain [28]. While these approaches offer enhanced versatility, they increase the overall complexity of the mechanical design. Furthermore, suspension systems improve the

overall vehicle's mobility by reducing incoming vibrations, making it possible to surpass obstacles more easily, while maintaining stability on challenging terrain. Independent or articulated suspension systems are increasingly being used by advanced autonomous vehicles. For UGV design, considerations for mobility must balance the mechanical complexity, adaptability to the terrain, and efficiency when handling obstacles in accordance with the given task at hand.

3.4 Hardware Design Components

The hardware configuration defines the physical characteristics, operational reliability, and performance of the UGV. From the suspension to the motor configuration and steering ability, these characteristics contribute to the grander goal, which is the vehicle's mobility, energy efficiency, and stability. This section outlines the key mechanical and electrical components that define UGV hardware design.

3.4.1 Steering and Wheel Types

The steering mechanism of the robot can be designed in two ways, skid steering or explicit steering [29]. Skid steering allows a vehicle to turn by independently controlling the wheel pairs on each side of the platform. Turning can be achieved in two distinct modes depending on the operating condition. At a standstill, the motors on one side rotate in the opposite direction to those on the other side, enabling rotation about the vertical axis. During forward motion, turning is accomplished by increasing or decreasing the rotational speed of the motors on one side relative to the other. Skid steering is comparatively simple to implement and requires no additional components compared to explicit steering, which makes it a common UGV design, particularly for tracked platforms.

Skid steering allows for different types of wheels to achieve desired behaviours [30]. Two common wheel types are conventional wheels and mecanum wheels. Conventional wheels are available in different sizes, patterns, and materials, and they can be mounted on different types of rims. This is useful in rough terrain where an appropriate size and pattern can be utilised. Mecanum wheels offer movement and rotation in confined spaces using rollers mounted diagonally on the rim. However, this is not suitable in rough terrain. Explicit steering is the design used in normal cars, where the wheels are rotated to make the vehicle perform a turning motion. This method requires extra components to control the angle of the wheels compared to a skid steer, but it requires less power to operate. Skid-steered vehicles require fewer components and are not as complex to design.

3.4.2 Motor Types and Placement

Several motor architectures exist for UGVs, each with distinct advantages depending on terrain, size, and control requirements. Brushed and brushless Direct Current (DC) motors are most commonly used in mobile robotics [31]. Despite being more expensive, brushless motors are more reliable, efficient, and have a higher torque density compared to brushed motors, which are simpler and more cost-effective, but have a shorter operational lifespan due to mechanical wear. More integrated configurations, such as in-hub motors that are mounted directly within the wheel, are being investigated by modern designs. This technique reduces mechanical transmission losses and enables precise independent control of each wheel, improving the drivetrain handling and efficiency. Dou & Zhou [32] demonstrate that hub-motor architectures that utilise multi-gear parallel reducers can enhance overall torque transmission, their load distribution, and stability control in UGVs. However, in-hub designs also increase unsprung mass, which can negatively affect suspension performance in rough terrain.

The design of the motor impacts the mounting options on the vehicle and its dynamics [33]. Motors mounted within the chassis and connected to the wheels via drive shafts centralise mass within the vehicle, resulting in lighter wheels with reduced rotational inertia, which facilitates steering. In contrast, hub motors add mass to the wheels, increasing inertia and making steering more challenging. At the same time, hub motors eliminate the need for drive shafts, simplifying the drivetrain, and may improve mass distribution by potentially lowering the centre of mass. Additionally, mounting the motor within the wheel rim does not negatively affect ground clearance, as the components are isolated from ground contact. Due to their effect on steering performance, wheel-mounted motors are uncommon in conventional vehicles that use explicit steering mechanisms.

3.4.3 Suspension Types

Suspension systems can be divided into two main categories: Independent suspension and dependent suspension systems [34]. These categories can further be divided into subcategories of different suspension designs. Each suspension system provides its own advantages and disadvantages and the choice of suspension system and design to use is therefore dependent on the vehicle's operational environment [35].

In a dependent suspension system, each wheel pair - for example, the front left and front right wheels - is physically connected. This configuration enables the system to translate movement and force experienced by one wheel to the other [36]. These types of systems lend themselves well to heavy-duty vehicles or vehicles that will see a high lateral load while turning. Such designs can improve full surface contact with all wheels. Common types of dependent suspension include:

1. *Leaf-sprung solid axle:* Consists of a fixed axle connecting each pair of wheels into a rigid body. For a driven axle, a differential is integrated into the axle housing. The axle is connected to the vehicle's chassis using leaf springs [37]. Leaf springs are constructed by stacking several steel plates of descending length. The springs in this system are used to fix the axle in the horizontal plane and support the vehicle load [38]. Leaf-sprung solid axles are among the simplest and earliest types of suspension systems.

2. *Four-link solid axle:* A four-link or multi-link solid axle uses rigid links to replace the leaf-spring. This fixes the axle in the horizontal plane. Coil springs are used to carry the load on the axle instead of leaf springs [39]. This results in a mechanically stiffer suspension layout, as the springs do not have to simultaneously locate the axle and dampen chassis motion. This construction adds complexity, as the geometry of the supporting links must be designed with specific geometric constraints to properly locate the axle.

In independent systems, each wheel is allowed to move freely, without any movement or force applied to the opposite wheel. The introduction of an independent suspension system is advantageous regarding comfort and stability. The benefits of this method are an important addition when it comes to the design of modern vehicles [40]. Independent suspension is therefore a common design used in passenger cars and all-terrain vehicles, as it improves wheel-ground contact when navigating challenging terrain or obstacles [34]. Common types of independent suspension include:

1. *MacPherson suspension:* It is one of the most common front suspension designs in modern vehicles [35]. This is due to the compact geometry and efficient use of space of the MacPherson suspension. This suspension design uses a single lower control arm (wishbone), mounted to the chassis and connected to a wheel spindle [41]. The spindle is also connected to a strut, which in turn is mounted to the chassis. When used as front suspension, the strut also acts as a steering point, allowing the wheel to be turned in either direction. The MacPherson is mechanically simple and suitable for applications where spatial constraints are present. A disadvantage of this design is that the spindle is fixed to the strut, the strut geometry induces changes in caster, toe, and camber angles when compressed.

2. *Double wishbone suspension:* This design is conceptually similar to MacPherson suspension, but with an added upper wishbone. The added wishbone acts as another anchoring point between the spindle and chassis, reducing the role of the strut as an anchoring component [42]. An added feature of this type of suspension is that the angle of the spindle can be maintained throughout suspension travel [35]. This design improves kinematic stability, where the caster, toe, and camber angles can be adjusted and kept throughout the suspension travel. A double wishbone design is therefore a suitable choice when a large amount of suspension travel is needed without changing the wheel alignment relative to the ground.

A visualisation of a dependent and an independent suspension system is presented in Figure 2.

3.4.4 Material Selection for Structural Components

A wide range of materials can be considered for the structural design of UGV, each associated with specific advantages and disadvantages. UGV's offer a huge variety of applications, designs, and weight classes, meaning that materials will vary. In practical UGV applications, metallic materials such as steel and aluminium are commonly used, while composite materials, such as carbon fibre, are rare and reserved for lightweight or high-performance designs. Steel provides high strength and durability, but with increased mass,



Figure 2: A visualisation of how dependent and independent suspension systems interact when faced with an obstacle on one side of the vehicle. The image provided in the left side is a representation of a dependent suspension system and the image provided in the right is a representation of an independent suspension system.

which may negatively affect vehicle mobility and energy efficiency. Aluminium is therefore a frequently selected structural material due to its strength-to-weight ratio. Even though aluminium is comparatively weaker than steel, its weight reduction is an important advantage [43, 44]. Carbon fibre composites offer even higher strength-to-weight ratios. However, the higher manufacturing complexity and cost, as well as considerations related to electrical conductivity, often limit their use to specialised applications rather than general-purpose UGV platforms [43, 45].

Carbon fibre composites are known to exhibit a comparatively brittle failure mode, where increasing fibre content compromises ductility and reduces energy dissipation during failure [46]. In contrast, aluminium alloys generally exhibit ductile behaviour, allowing for deformation prior to fracturing, which is advantageous in impact scenarios [47]. When combined within the same structure, these materials complement each other: carbon fibre provides high stiffness and strength, while aluminium contributes ductility and higher impact tolerance.

3.4.5 Battery Selection

In mobile robotics, including UGVs, batteries serve as a primary power source for all of the electrical components [48]. The battery capacity is expressed in ampere-hours (Ah) and indicates the total amount of charge the battery can deliver, while the voltage (V) represents the electrical potential at which charge is delivered. Lithium-ion (Li-ion) batteries [49] are the most widely used type of rechargeable batteries and are the leading type in electrified transportation, due to their high energy density, long life span, and low maintenance requirements.

Lithium-ion polymer (LiPo) [50] batteries are a type of Li-ion battery that uses a gel-like polymer electrolyte instead of the liquid found in conventional Li-ion batteries. This solution allows for manufacturing in various shapes and sizes, which is one of the reasons they are used in many portable electronic devices. Another advantage of LiPo batteries is their high discharge rate. These characteristics make LiPo batteries ideal for applications that require long operating times, with high peak current demands, and limited weight requirements.

To ensure safe operation and optimal performance, lithium-based batteries require a Battery Management System (BMS) [51]. The BMS monitors critical parameters such as voltage, current, temperature, and state of charge. It protects the battery from potential issues like overcurrent during charge and discharge, reducing the risks of overheating and damage. Another crucial task is battery balancing, which ensures an even charge distribution across the battery cells. Lastly, the BMS contributes to thermal management by monitoring temperature and enabling protective measures when necessary. This is an important functionality that reduces the risk of thermal runaway and ensures the battery operates in safe conditions.

Regenerative energy recovery is a common technique used in electric drive systems, where kinetic energy generated during braking or freewheeling is converted into electrical energy stored in the battery system [52]. By redirecting this energy instead of dissipating it as heat, regenerative systems improve the overall energy efficiency and extend operational runtime while simultaneously acting as a secondary source of charging. This will, in addition, reduce thermal losses in power electronics and braking components, contributing to overall system temperature and improved reliability.

3.5 Software Architecture

The software framework has had a significant impact on the development of UGVs. Many earlier platforms relied on custom-built microcontroller systems that offered rudimentary control, but had limited integration and scalability with different sensors and actuators [53]. Teleoperation remains a crucial component of earlier and modern UGV systems. This technology provides a reliable means of manual control suitable for specific scenarios, such as during system testing, deployment in challenging terrain, and safety-critical situations where autonomous behaviour may be insufficient or unreliable. Furthermore, teleoperation serves as an essential fallback mechanism, allowing human operators to intervene when sensor data is degrading or unexpected obstacles occur [54]. Middleware frameworks were developed to increase modularity and interoperability as the demand for autonomy increased. ROS2 emerged as a leading solution because of a large open-source ecosystem of reusable packages and established interfaces for motion control and sensor integration mapping. By increasing support for many sensors and navigation systems, including Nav2 and SLAM Toolbox [55], and enhancing real-time capabilities, quality of service, and communication reliability, ROS2 has addressed the limitations of the original framework. The design of the UGV software has changed from inflexible platform-dependent systems to modular architectures that enable quick development and modification across a wide range of applications.

3.5.1 Navigation and Teleoperation

Navigation and autonomy continue to present central challenges in UGV development. Earlier systems were limited in their flexibility and were operated in controlled environments, since they could only teleoperate directly or follow predetermined paths [11]. Modern platforms are more sophisticated, where they increasingly rely on adaptive path planning, probabilistic localisation, and simultaneous localisation and mapping to function well in obscured or not well-defined areas [4, 56].

Furthermore, modern platforms also incorporate sensor fusion techniques to compensate for the shortcomings of individual sensing modalities, such as limited perception in adverse weather, signal degradation, and occlusions, affecting Light Detection and Ranging (LiDAR), Global Positioning System (GPS), and camera-based systems. Recent advancements in data-driven perception and machine learning allow UGVs to operate efficiently in unpredictable environments, improving their ability to recognise and classify different objects while allowing them to adjust to changing conditions [4]. Despite this, real-time computation, communication limitations, and the inherent unpredictability of outdoor environments make even these sophisticated autonomy systems encounter performance degradation.

4 Related Work

This section reviews existing research on UGV design, focusing on structural integrity, mobility, autonomy, modularity, and integral design choices. These research areas collectively contribute to the overall systems performance, adaptability, and operational sustainability of UGV platforms and inform the design choices of this project.

4.1 Modularity and Reconfigurability

Modularity is another key principle in UGV design, enabling parts to be swapped or reconfigured without rebuilding the platform. Gadekar et al. [57] demonstrate how modular chassis and sensor payloads increase adaptability across missions, while Ćosić et al. [58] demonstrates how a single UGV can be equipped with interchangeable tool modules for specialised tasks such as mine detection or debris removal. Lyness [22] emphasises that modularity introduces both weight and complexity, making it a big consideration in how to design this to comply with the requirements in Table I. Modular mounting points and a lightweight chassis are therefore essential to ensure adaptability, enable long-term maintainability for future work, and secure compliance with ARES Makeathon requirements. These findings reinforce the need for a lightweight yet structurally robust chassis that can accommodate reconfiguration without compromising performance.

4.2 Mobility and Locomotion

Wheeled locomotion remains a prevalent solution for agricultural UGVs due to its mechanical simplicity and efficiency in moderately challenging terrain. German et al. [59] presented a four-wheel design with independently driven wheels, off-road tyres, and a vertical suspension system capable of maintaining ground contact while traversing challenging terrain. This form of suspension-based adaptability is directly relevant to this project, as similar mechanisms are considered to improve mobility on challenging terrain in this project. Farella et al. [13] survey multiple wheel-based systems, including the "X-bot" robot [60], which employs a four-wheel drive (4WD) configuration with tractor-style tires optimized for muddy environments. Several commercially available UGVs on the market employ a similar design, one example is the Summit platform developed by Robotnik [61]. The summit is designed using four 500 W hub motors to enable independent drive of all wheels in combination with independent suspension. This approach is desirable for this project; however, implementing a suspension system of similar complexity would require substantially more development and time compared to using commercially available off-the-shelf solutions. The Summit UGV has a total weight of 65 kg and a payload of up to 50 kg, which is similar to the platform developed in this project. The usage of 500 W motors can therefore be used as a guideline in this project, though the torque, rpm, and other specifications of Summit's motors are unknown.

Parillat et al. [62] propose an alternative approach to navigation in rough terrain by introducing a variable-geometry single-tracked vehicle capable of adjusting its track tension to conform around obstacles. By decreasing the track tension, the contact area between the track and terrain increases, enhancing both traction and adaptability on challenging terrain. This idea illustrates how sophisticated tracked systems can remain stable and mobile in extremely erratic terrain such as loose soil, rocky terrain, or debris-covered surfaces. As a result, such systems are appropriate for certain uses where more traditional tracked or wheeled vehicles would find it difficult. However, while such designs offer impressive off-road performance, their mechanical complexity and weight make them less suitable for this platform, which emphasises robustness, modularity, and maintainability. If a tracked configuration were chosen, a dynamic concept as proposed by Parillat et al. could have been explored further, particularly given its advantage in the aforementioned types of terrain, given that it is within the weight requirement limit. Nevertheless, considering the project's requirement and time constraints, the rapid prototyping of a dynamic tracked system is deemed out of scope due to its complexity.

In the early stages of prototyping, it is often impractical to manufacture using the final intended material due to cost, availability, and manufacturing complexity. Instead, simpler and more accessible materials such as plastic or wood are commonly used to enable rapid prototyping and iterative design evaluation. Xing demonstrated this approach by constructing the initial prototype using wood, taking advantage of its flexibility [44]. This method allowed for efficient evaluation of geometry, layout, and functional concepts before transitioning to more advanced materials close to the final product.

4.3 Suspension

The mobility and operational reliability of a UGV are influenced by suspension systems [63]. Independent suspension has been shown to reduce vibrations that negatively affect sensor performance, while also improving obstacle crossing and terrain stability [64]. Zou et al. demonstrate a sophisticated configuration, such as six-wheel drive combined with four-wheel steering, to further improve manoeuvrability in challenging terrain [65]. Traversing difficult terrain is an important aspect for many UGVs, as operational environments vary significantly. Consequently, suspension designs are often tailored to the specific terrain and obstacles expected during operation. For example, Althani et al. employ a double bogie system which is used to optimise the UGVs performance in desert environments [66]. Their design consists of a traditional longitudinal bogie system at the front and a single transverse bogie in the rear, providing improved load distribution. While a bogie-system aid in load distribution, these systems are usually more suitable for heavier loads due to increased complexity and mass. Platforms that are intended to traverse severely undulated ground benefit from such suspension systems. Furthermore, a bogie system is also not suitable for a platform consisting of two wheel-pairs, as a bogie system connects the wheel-pairs to a single pivot point, adding complexity. Another important consideration is that some obstacles cannot be navigated around. In such cases, passive suspension systems may be insufficient. One approach to address these scenarios is to have an active articulated suspension system, which allows the vehicle to raise or lower individual suspension elements to maintain ground contact while climbing obstacles [67]. Attia [68] utilises linear actuators to actively control suspension geometry, enabling adjustments to the centre of gravity, improving platform stability when navigating challenging terrain. Similarly, Zhang et al. demonstrate a multi-wheeled UGV that employs actuators to dynamically alter the suspension's geometry, allowing obstacle traversal without significantly altering the UGV's body orientation [69]. These studies highlight the importance of suspension design for maintaining mobility and operational reliability in challenging terrain. Passive suspension systems are often adopted as a compromise between mechanical simplicity and reliability, depending on their intended operational use. Actuated systems increase the overall performance, but at the expense of increased cost, weight, and complexity.

4.4 Propulsion

Propulsion technologies for UGVs vary significantly depending on operational requirements and application domains. Depending on the propulsion choice, factors such as weight, noise, and maintenance are affected. Gasoline engines offer great power and long endurance, while also increasing mechanical complexity. Hybrid approaches combine electric and combustion systems to increase range [70]. Although battery capacity limits them, electric drives are lighter, quieter, and more effective. Since battery placement directly impacts the centre of gravity and slope stability due to its weight, it has been recognised as a crucial component in UGV design. Ghobadpour & Co. draw attention to a similar trend in agricultural UGVs where electrification increases controllability and efficiency while hybrids increase range at the expense of complexity and weight [71]. In light of these conclusions, the project uses an entirely battery-powered electric drivetrain, putting efficiency and controllability ahead of longer range.

4.5 Stability and Structural Considerations

Maintaining stability on inclines is a significant concern for UGVs. Tao et al. investigated stair-climbing in tracked robots and demonstrated that the centre of mass concerning the track support polygon significantly affects safe ascent [72]. Experiments indicate that a rearward centre of gravity increases the likelihood of tipping and failure in climbing. The same concept applies to sloped surfaces, despite its focus on stairs: a greater or rearward mass distribution decreases the highest safe slope angle. Besides mass distribution, stability is affected by geometric factors such as wheelbase and track width. Batteries represent a significant portion of the overall mass, so their arrangement, along with the positioning of other heavy parts, is a crucial element in meeting the 30° incline requirement of the project.

4.6 Sensor Placement and Perception

UGV platform design relies heavily on sensor placement because improper placement can limit perception and decrease overall data quality. The value of complementary modalities was highlighted by Rawashdeh and Jasim [73] who showed that integrating LiDAR GPS cameras and ultrasonic data enhances navigation

reliability. Their work emphasises the significance of physical placement on the chassis to maximise coverage and minimise occlusions from the chassis, even though it focuses on sensor fusion. Furthermore, vibrations from the traversal can skew depth perception and overall data quality for all sensors [63]. Suspension can help reduce vibrations, ensuring improvement in sensor data quality, while reducing overall deterioration of the platform. GNSS antennas need to be placed in an elevated, unobstructed location to improve signal quality. Standardised mounting points at different positions and heights are therefore essential to ensure reliable perception and ease of future sensor integration.

4.7 Navigation, Teleoperation, and Autonomy Frameworks

UGV operation can be realised at different levels of autonomy, ranging from direct teleoperation to fully autonomous navigation. Horeličan et al. demonstrated that the ROS2 Nav2 stack enables meaningful autonomous navigation on custom platforms, supporting localisation, planning, and sensor integration in semi-structured environments [56]. Teleoperation, on the contrary, remains essential when communication is unreliable or environments are highly uncertain. Moniruzzaman et al. review teleoperation methods, highlighting challenges of latency, control lag, and reduced situational awareness, while proposing assistive overlays and predictive control as mitigation strategies [74]. Cieślik et al. further show that camera placement significantly affects operator performance in teleoperated UGVs, directly linking sensor configuration to control effectiveness [75]. Mixed-initiative approaches combine these modes: Chiou et al. present a framework where control shifts dynamically between teleoperation, supervised autonomy, and full autonomy depending on task and context [76]. Surveys of ROS2 adoption confirm that the middleware provides the modularity and quality-of-service features required to support such variable autonomy levels, and will therefore be used in this project for the teleoperation implementation [77].

5 Method

This section describes the project phases and the methods used to achieve the expected outcomes. An initial plan established the foundation for the project by identifying key subsystems, design constraints, and team dynamics required for successful development. Defining the platform's functionality at a high level was a necessary step in order to increase the probability of delivering a suitable platform that aligns with the team's intended goals. At the beginning of the project, a comprehensive risk assessment was conducted to identify and visualise potential issues that could arise during the project's duration. This assessment served as a preventative measure to mitigate risks and ensure preparedness for unforeseen challenges. The working process, together with the methods used in each phase, is illustrated in Figure 3. The following subsections describe how the project methods were utilised in each phase.

5.1 Literature Review & Market Research

After planning the project's progression loosely, a deeper research phase began. The project could delve into two major directions: purchasing a pre-built platform, and modifying it in accordance with the requirements listed in Table I, or building a platform from scratch, each with its benefits and drawbacks. The two directions resulted in research of two different areas, both a literature review and market research regarding complete platforms as well as components to build one. The first option offered potential time savings and reduced development complexity, as it would allow the team to work from an established and tested base. However, it also presented limitations in terms of adaptability, cost, and compliance with fulfilling all aforementioned requirements. Conversely, the latter option would provide greater design flexibility and the opportunity to design the system for the given requirements, though at the cost of increased development time and increased technical risk due to inexperience and time constraints. Ultimately, the next step was to decide which direction to take. All data from the research was gathered, and the potential candidates for purchasing a platform were compared with each other. When the project direction settled for building a UGV from scratch, proposed solutions were presented and implemented.

5.2 Design and Implementation

After the literature review and market research as a foundation, the project transitioned into the design aspect in development. This design process involved integrating mechanical aspects, electrical and software subsystems into a cohesive architecture while ensuring compliance with the project requirement listed in Table I.

5.2.1 System Design

Building the UGV platform required the integration of multiple mechanical, electrical, and software subsystems into a cohesive and functional system. This section provides the main implementation choices related to platform construction and subsystem integration. The designing phase began with establishing and structuring an overview of the subsystems required to build a functional and complete UGV platform. To obtain a comprehensive overview of all subsystems within the platform and ensure that all relevant aspects were considered in the design process, two complementary methods were applied: a conceptual system map and a system architecture mapping. These systems provide a structured representation of subsystems and their interdependencies within the overall system, helping to clarify design priorities, identify potential issues, and reduce uncertainty during implementation. Together, these visual diagrams form the methodological foundation used throughout the project.

5.2.2 Hardware

The structural components were designed using SolidWorks [78], a 3D computer-aided design (CAD) software. The CAD models were used as a design basis and to provide blueprints for manufacturing. Moreover, welding was used to permanently assemble structural components. The design of the electrical system was created in Microsoft Visio [79]. The Power Distribution Board PDB was designed in KiCad [80], an open source software used for designing electronic schematics and Printed Circuit Board (PCB). Manufacturing of the PDB was made by an external manufacturer. The components was hand soldered on to the PCBs.



Figure 3: The figure illustrates an overview of the project workflow. Each color-code is associated with a specific method and its corresponding tools and techniques. The working process began with stating the research problem. After that the team began to set up a plan for how the project should proceed. The following phase included a literature review and a market research, resulting in a proposed solution. The next step was an iterative phase were implementation, testing and evaluation would either lead to re-design or validation against the requirements. Lastly the process and results were documented both as a technical report but also for handover purposes.

5.2.3 Software

The software designed in this project included the motor control, battery behaviour, safety measures, and the teleoperation of the UGV. This software system was designed using ROS2 [81] as the underlying operating framework, with Python and C++ employed as the primary programming languages. Python for high-level testing of motor control and battery-related functionality and safety, as well as familiarisation with the ROS2 environment. C++ was selected for the main software development due to its closer interaction with hardware and better performance characteristics.

5.3 Test and Analysis

Throughout the project, testing, evaluation, and lastly validation were made to ensure the proposed solutions met the project's expected outcome. Testing verified the correctness of each component. The tests were functional but also stress tests to explore component limits. Evaluation assessed the quality of the solutions, and validation was made through comparing the results with the pre-defined requirements.

5.4 Documentation

To support future operation of the UGV and handover, a structured user manual was produced. The document covers all the included parts and components in the platform, all platform features, and how to use the platform. It also includes safety instructions and regulations. The user manual was written in markdown files and uploaded to a Git repository [82] for this specific project. The project was lastly documented into a technical report in Overleaf using LaTeX [83, 84].

6 Ethical and Societal Considerations

The development and implementation of UGVs involves a variety of ethical, social, and legal issues, in addition to operational and technical difficulties. Apprehensions regarding the influence of robotics technologies on issues such as employment, environmental sustainability, and human responsibility are raised by their growing integration into both civilian and military applications. The ethical and societal aspects of UGVs autonomous systems and AI are critically examined in this section, particularly in military domains, given the project's defence affiliation.

6.1 Defence Context and Ethical Positioning

The Swedish Defence Materiel Administration (FMV), the national organisation in charge of purchasing equipment and technical systems for the Swedish Armed Forces, is associated with this project. Additionally, it is carried out within the framework of the ARES Makeathon, an European Defence Agency (EDA) initiative that encourages research within the military sector, places the project in a defence-oriented context and emphasises the use of UGVs. The underlying technologies are applicable to both military applications such as reconnaissance, logistics support, and operational assistance, and civilian domains such as infrastructure inspection, agriculture, and disaster response. This project is intended for educational and research purposes. Although there are fewer direct ethical risks when working in an academic competition, the project nevertheless contributes to discussions about technologies that may have an impact on future adoption by the military and civilian sectors.

6.2 Economic Implications of Automation

From an economic perspective, productivity growth and labor market structural change are directly related to the integration of UGVs and automation in general. According to Graetz and Michaels [85], the adoption of robots has greatly increased output per worker, adding roughly 0.36 percent to labor-productivity growth annually across 17 advanced economies, while also lowering production costs and output prices. Overall employment effects appear to be modest despite the uneven distributional impact. Between 1980 and 2016, low-risk occupations saw an 18 % increase in employment while high-risk automation-prone occupations saw a 6 % increase. This implies a redistribution of work instead of a total reduction [86]. Regional effects are disruptive, according to research on the American labor market. For every industrial robot added per 1000 workers, the employment-to-population ratio in affected areas dropped by 0.18 to 0.34 percent and average wages by 0.25 to 0.5 percent. Manufacturing-intensive areas were most affected [87, 88]. These findings demonstrate how automation can exacerbate inequality: highly skilled workers benefit from the new technical roles, while low-wage workers in routine or service jobs are disproportionately at risk of displacement. Complementary in higher-skill roles, such as system integration, supervision, and maintenance, as well as substitution in hazardous or repetitive field tasks in industries such as transportation and logistics, are likely to influence the economic impact on UGVs. Additionally, the risks of displacing low-wage service workers outweigh the efficiency gains in service industries like retail, healthcare, and urban logistics, where they are being tested more frequently. The outcome is determined by how well programmers are retrained, and how much increased productivity creates new demand in related industries. Projections of labor market trends reflect this duality: routine manual jobs are expected to grow slowly or decline, while technical and engineering jobs related to the development and maintenance of automated systems are expected to rise [89, 90].

6.3 Legal and Regulatory Perspectives

The development of UGVs brings up significant legal issues, especially when it comes to defence situations where international humanitarian law is applicable. The concepts of distinction between (combatants and civilians) and proportionality (ensuring that civilian harm is not excessive in relation to military advantage) are crucial to this. Since fully autonomous systems are currently unable to consistently ensure adherence to these principles, there has been substantial legal discussion about their legitimacy in armed conflict [91, 3]. Therefore, whether new treaties or legally binding regulations are needed for autonomous weapons such as UGVs is still being discussed internationally at the United Nations Convention on Certain Conventional Weapons. UGVs must also abide by laws in civilian settings, such as those pertaining to product safety and liability. Whether the operator, the manufacturer, or the system integrator should be held accountable in

the event of a malfunction or accident raises questions. When autonomous or semi-autonomous machines are used in public or commercial settings, the European product safety law and the upcoming EU Artificial Intelligence Act emphasise the necessity of distinct lines of accountability [92, 93]. Data protection and privacy are similar issues, particularly when UGVs use cameras, LiDAR, or other sensors to navigate and gather environmental data [86].

6.4 Environmental and Sustainability Considerations

There are significant sectoral and environmental issues raised by UGVs. Robotic platforms can lessen environmental pressures in certain applications, according to life-cycle assessments. For instance, compared to conventional methods, autonomous weeding robots in agriculture significantly reduce greenhouse gas emissions and pesticide use [94]. Nevertheless, the material intensity of robotics offsets these benefits. Batteries, semiconductors, and rare-earth minerals have environmental costs associated with their production and disposal, particularly if recycling and reuse pathways are unavailable [95]. Automation can increase productivity in manufacturing and logistics outside of agriculture by reducing energy waste and optimising resource use [96]. However, there is a risk associated with rebound effects because increased efficiency may also increase the demand for energy and accelerate material consumption, undermining sustainability goals. Almusharraf A. I. argues that circular economy frameworks should incorporate robotics, highlighting the importance of modularity, repairability, and ethical sourcing [97] Incorporating such practices is in line with EU sustainability initiatives, guaranteeing that robotics development supports long-term climate and resource goals, and that efficiency benefits are not outweighed by ecological costs. In the future, the development of UGVs will be influenced by both technological advancement and governance: whether or not UGVs are embraced as trustworthy, sustainable, and responsible technologies will ultimately depend on international legislation, regulations, and societal expectations [98].

7 Implementation

This section presents the implementation of the UGV platform based on the design decisions and system-level planning described in Section 5. The implemented system architecture, as well as the drivetrain, electrical subsystems, and software communication, are described. Details regarding component selection, mechanical design, and assembly are then presented. Finally, the testing phase and its execution are described.

7.1 Project Management

A Work Breakdown Structure (WBS) is a commonly used project management tool for structuring complex projects by decomposing the overall scope into smaller, manageable components, both in smaller and larger groups. By organising work into hierarchical levels, a WBS supports planning, responsibility allocation, and progress tracking throughout a project. Based on the WBS, Gantt charts, a methodology created by Henry Gantt [99], are frequently used to visualise project schedules by assigning start and end dates to activities; thereby enabling the identification of task dependencies, time allocation, and critical moments

A WBS was developed during the beginning of the project and is presented in Appendix 1.2. During this time the direction of the project was to by a pre-built platform. Although the WBS are not reflecting the individual task the project actually included, it still provides the main structure and helped the team to align with the overall deadlines. The work done in the project was divided into five parts. Planning, including risk analysis and research, the developing time frame included design and implementation of the platform. Moving on to the test phase all the requirements were to be validated. The integration phase included collaboration with the parallel sub group working on the autonomy. Lastly, the project should end in a smooth handover phase. The corresponding Gantt Chart is visualised in Appendix 1.1 and consists in the same phases as the WBS. The time allocated for each task is shown trough the purple horizontal bars. The left and of the bar indicates the tasks start date and the right indicates the end date. Sequential bars show how tasks must be completed in order.

7.2 Research Phase

The project began with one key question in mind: Whether to build a platform from scratch or purchase an existing platform and modify it to meet the requirements. The answer was found after a one-month planning and research phase.

7.2.1 Purchase a Pre-Built Platform

During the initial stages of the project, the main objective was to do a market analysis on the existing pre-built platforms, as doing so would significantly reduce development time. A total of 20 potential platforms were analysed through a structured risk assessment framework designed to evaluate their suitability for this project. Each platform was then rated according to its adaptability to the project requirements specified in Table I, along with other factors such as lead time, drivetrain, and country of origin. Based on this analysis, the two most promising candidates were selected for a more detailed evaluation.

The Robotnik Summit [61], was one of these top candidates, however, its total weight posed a significant challenge. Replacement of the wheel and battery with lighter alternatives could potentially bring the platform within the required range of R1 in Table I. However, this modification would effectively only retain the chassis, suspension and motors of the original design, since the sensors and computing units onboard would also be replaced. Therefore, a decision was made that changing the platform was not justifiable in regard to the base platform's high cost. Consequently, the Robotnik Summit platform was excluded from further consideration, but was used as a reference during the design of the UGV developed in this project.

Robodyne's MAXXII STANDARD [100] was another top contender. Despite its initially favoured specification and lower pricing relative to the previous platform, its tracked drivetrain configuration was considered unsuitable for the intended application. The resulting stiffness from the lack of suspension would cause excessive vibration, negatively affecting the sensor's performance. For this reason, the MAXXII STANDARD platform was also excluded. R5 in Table I proved to be a deal-breaker for most tracked platforms available on the market, since the majority of them lacked a suspension system within the weight class of the study. Following these evaluations, the project team and stakeholders decided to develop a platform from scratch.

TABLE II
COMPARISON OF PLATFORM SPECIFICATIONS

Platform	Cost (kr)	Weight	Dimensions	Gradeability	Runtime	Max Speed	ROS2	Drive Train	Country	Lead Time
MAXXII Standard	115 255	60 kg	181.2 cm	30°	4 h	1 m/s	Yes	Track	Italy	1–2 months
Robotnik-Summit	330 000	65 kg	175 cm	36°	5 h	3 m/s	Yes	4WD	Spain	1 month

Table II compares the prebuilt MAXXII Standard and Robotnik Summit platforms against each other based on important parameters set by the project group. A full-scale risk analysis between these two platforms was evaluated and listed in Appendix D.

7.2.2 Market Research

The market research showed that purchasing a pre-built platform would not be cost-effective, so the research for building a platform from scratch began. To maintain modularity throughout the platform, the goal was to use as many off-the-shelf components as possible, simplifying future modifications and reducing overall development time.

The market research provided valuable insights into which features a platform should include. After evaluating the advantages and disadvantages of tracks and wheels as mobility systems for the UGV, wheels were ultimately selected. Tracked systems offer arguably better traction and stability in rough terrain, but are mechanically more complex, heavier, and more difficult to maintain. In contrast, wheeled configurations provide simpler construction, lower rolling resistance, easier maintenance, and a wider selection of commercially available supply. Additionally, the tyres contribute a natural damping effect, reducing vibrations, and the integration of a suspension system is easier to implement than on tracked vehicles. Furthermore, wheels are quicker and cheaper to replace if damaged during testing or operation, which improves maintainability and reduces long-term costs.

Based on the evaluation of the most relevant platform on the market, a similar in-wheel motor configuration was selected.

7.3 Design and Manufacturing process

To support the implementation of both the electrical and mechanical subsystems, a conceptual system map and system architecture were implemented to provide an overview of the platform components and their functions and interdependencies. The overall physical components and their structural organisation and their interactions, together with the software components and their interactions through communication pathways, are illustrated in Figure 4. In parallel, a mechanical system map was created to visualise the structural layout of the platform. The chassis functions as the primary load-bearing structure, integrating the suspension, motors, wheels, and shock absorbers, while also illustrating how electricity is distributed to these subsystems. Together, these system maps enabled a structured implementation process and facilitated effective subsystem integration and overall, as well as providing awareness of potential risks prior to implementation. The system map diagram is presented in Figure 5.

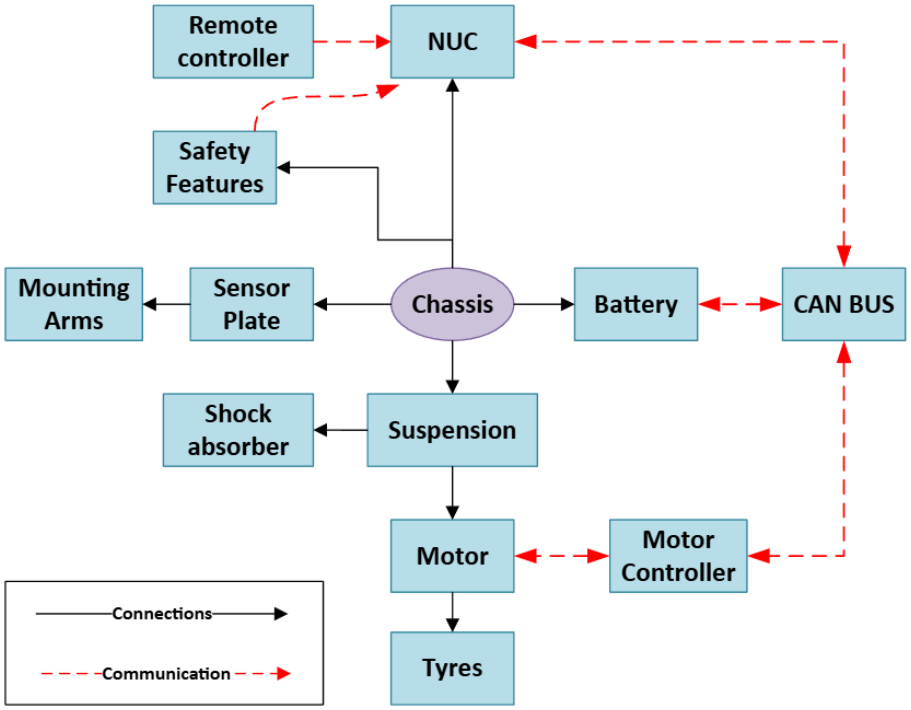


Figure 4: The system map was done with the intentions to give the group a common overview of how all the parts in the system connects together. Creating the map helped to identify impotent dependencies between different parts of the UGV, making it easier to communicate about the task distribution among the project group. The red arrows describes the dependences for components that needs communication with each other. The black arrows illustrates the dependencies for the physical components included in the chassis.

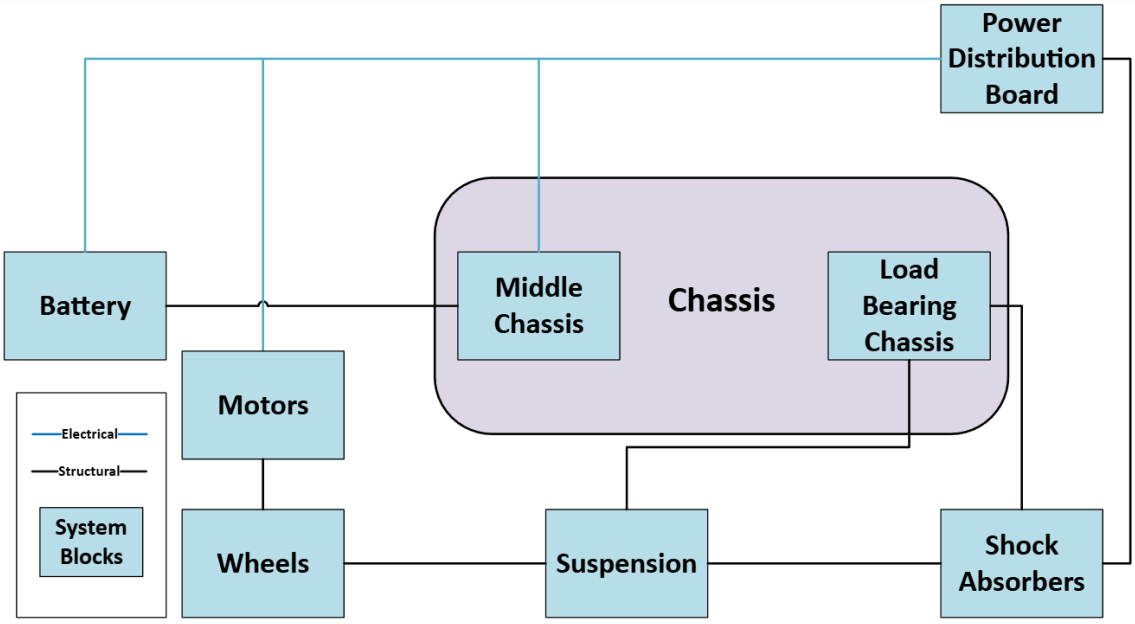
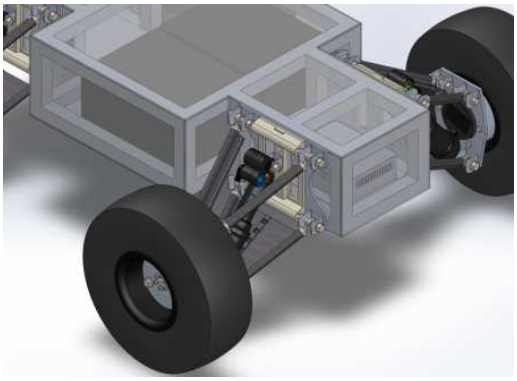


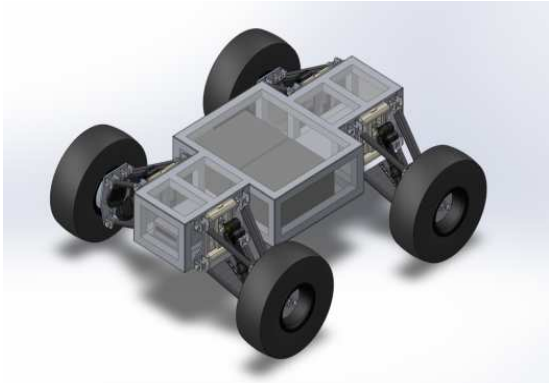
Figure 5: This system architecture was designed to visualise the mechanical aspects of the platform, and all of their structural and dynamic subsystems interact. This mapping methodology helps identify load paths, spatial constraints, relations between components, and was used as preparation before doing major design choices. The middle-chassis acts as the middle compartment of the UGV housing the battery, and the load-bearing chassis is the front and rear compartment of the UGV holding the suspension solution, motors, and wheels. The front and rear are connected by two aluminium beams extending the full length of the platform, and can be seen as a separate load-bearing part of the chassis.

7.3.1 Prototyping

The initial design was created in SolidWorks [78] and 3D-printed in Polyactic Acid (PLA) to validate different geometries for the suspension and mitigate the risk of parts colliding with one another. This also provided an overview of how the suspension would behave during suspension compression. The final prototype design was exported from the design software to blueprints, shown in Figure 6.



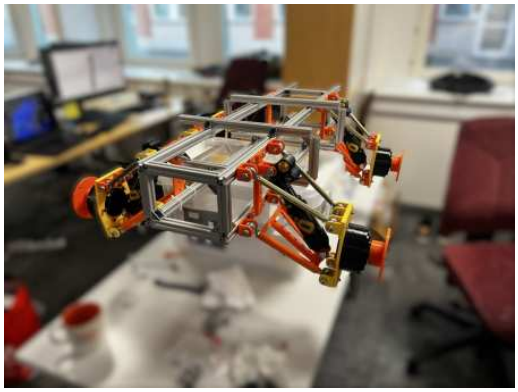
6.1. Blueprint zooming in at the suspension solution.



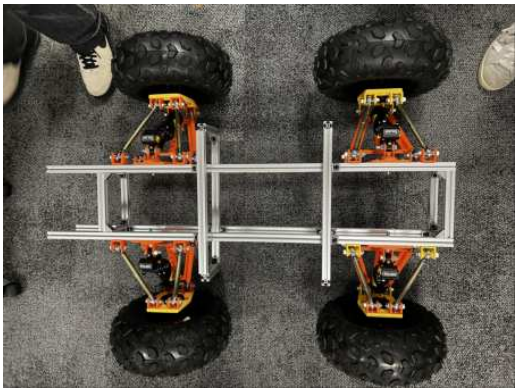
6.2. Blueprint for the complete UGV.

Figure 6: Schematic in SolidWorks that served as a design blueprint before manufacturing the chassis.

Since the initial prototypes were made with 3D-printed parts, any necessary changes to the models were easily made, and a new version was quickly available for testing. During the prototyping phase, 3D-printed parts were produced for one of the four suspension corners, serving as a representative corner to reduce print time. This approach allowed the design changes to be evaluated by printing a single component instead of all four, as further iterations were anticipated. The prototype frame was constructed using 20 mm x 20 mm aluminium profiles and designed exclusively with 90-degree angles to maintain simplicity and structural stability. Once the final design was decided, all remaining suspension components were 3D-printed to allow assembly of the suspension on the prototype frame. This enabled verification of the final dimensions, assessment of the overall platform size, and identification of any overlooked design aspects. Simplified mock-ups, such as cardboard models representing the battery and other major components, were used to validate packaging constraints, and the physical onboard computer was placed in the chassis to confirm fit and accessibility. The prototype is shown in Figure 7.



7.1. The prototype and all of the 3D-printed temporary solutions from a diagonal view.



7.2. The prototype and all of the 3D-printed temporary solutions from a view above.

Figure 7: The finalised prototype before manufacturing the chassis and all of the minor parts.

7.3.2 Chassis

A manufacturer then welded the chassis, which consists of a rectangular aluminium space frame manufactured from 20 mm x 20 mm square tubing, forming a rigid structural backbone for the vehicle. The frame is divided into multiple compartments, allowing for modular integration of component housing, such as

electronics, batteries, and the computer. Aluminium was selected for the load-bearing frame due to its high strength-to-weight ratio, while also providing sufficient rigidity and environmental protection. Manufacturing the frame using carbon fibre tubing would be possible, however, carbon fibre is a brittle material, while aluminium is ductile. This means that a frame made out of carbon fibre may crack, snap, or be punctured on impact or in a crash. Since aluminium is ductile, on impact in the case of a crash, the material may bend and be deformed, but still has the chance of maintaining cohesion. The central section of the chassis is wider to accommodate core systems and maintain structural rigidity, while the front and rear sections taper inward where the suspension assemblies are mounted. Carbon fibre composite panels were used as non-load bearing cover panels for the walls, floors, and roof of the chassis. The carbon fibre panels, which cover parts that may need access, were screwed on, while panels covering parts where access is not needed were glued on using silicone. An old donated carbon fibre sheet was measured, cut down, and reused to make all the cover panels for the platform. This geometry allows for longer suspension control arms, improving wheel travel and ground clearance without increasing the vehicle's overall width. Additionally, making the ends larger would obstruct the vision of the ground close to the robot for the sensors. Inside the central chassis, the batteries are placed to help control where the centre of mass is. The front section holds the electrical components, and the rear section holds the computer and router.

7.3.3 Wheels

The wheels of the platform are designed for small-sized all-terrain vehicles and have a diameter of 33 cm and a width of 14 cm. The tread of the tyres is suitable for rough terrain by ensuring traction on uneven surfaces. The tyres require an air-filled tube, and the pressure can be adapted to contribute to vibration damping and is mounted on each wheel. A critical consideration for the selected wheel design is a minimum inner rim diameter sufficient to accommodate the in-hub motor assembly. The chosen wheels satisfy this constraint, allowing the CubeMars motors to be mounted concentrically within the rim without interference or extensive overhang. Furthermore, the wheels provide the possibility to allow the motors to be mounted within the rim. The motors and rims are fastened using a custom-designed aluminium component and M8 bolts and nuts. The wheels were commercially available and can easily be replaced [101, 102].

7.3.4 Suspension

A double wishbone suspension, with adjustable linkage rods, was used to ensure that the angle of the wheel remains unchanged during the suspension travel. The adjustable rods also enable some length adjustments to be made and simplify wheel alignment.

A pneumatic mountain bike (MTB) shock-absorber provided by Öhlins Racing [103] is attached between the lower wishbone and an aluminium brace located between the upper and lower parts of the chassis. The shock is modified by Öhlins to lower the effective spring pressure and decrease the force to compress the shock-absorber. Since a single one of the provided shock-absorber is meant to hold and be compressed by the weight of a fully grown person. The modification of the shock-absorbers was needed to enable the low unsprung mass of the UGV, compared to the intended mass within the shock-absorbers operating window. The modification is done by drilling a small relief-hole in the pneumatic spring. The resulting air leak within the shock-absorber results in a lowered spring rate. A result of this modification is that the lower spring pressure is not enough to extend the shock-absorber on its own. To compensate for the loss of rebound, the internal chamber of the pneumatic spring has been modified to accept a standard coil-spring. The added standard coil-spring does not add to the effective spring pressure during compression, in a meaningful way, but compensates for the loss of rebound. With the modifications to the shock-absorbers, the operating window has been lowered to match the unsprung mass of the UGV. Then the ends of each wishbone are connected to an aluminium bracket that holds the electric motor. Two 150 mm linkage rods, with a right-hand and left-hand M8 internal thread, are used to construct the upper wishbone. The linkage rods are connected to brackets fastened to the chassis using uniballs. This allows free movement in the vertical plane and the ability to adjust the angle between each linkage rod and the centreline of the wishbone. To increase the rigidity of the suspension system and withstand the force applied by the shock absorber, the lower wishbone was designed as a single piece. Carbon fibre was chosen as a material for the lower wishbone because of its stiffness, together with its strength-to-weight characteristics. The component was designed and manufactured by the authors of this study. The lower wishbone was mounted to the chassis, similar to the upper wishbone.

The final design of the lower wishbone was concluded after testing geometry using link arms and uniball joints. The component was then constructed using unidirectional prepreg carbon fibre, Toray T700,

Divinycell® H80 core material, and woven carbon fibre fabric for the outer layers. Four wishbones were constructed bottom up, starting with a woven carbon fibre fabric. A unidirectional layer was placed on top with the fibres directed at an angle of 45 degrees to the centreline of the symmetric design, and another layer was placed with an angle of -45 degrees. Before repeating the procedure in a backwards manner, the core material was placed together with two woven carbon fibre tubes [104]. This resulted in a seven-layer component. The wishbones were then hardened in an oven for several hours before the threaded inserts were glued into the carbon fibre tubes. To finalise the components, holes were drilled on the centreline to allow for the shock absorber to be connected to the wishbone. The finished product is shown in Figure 8.

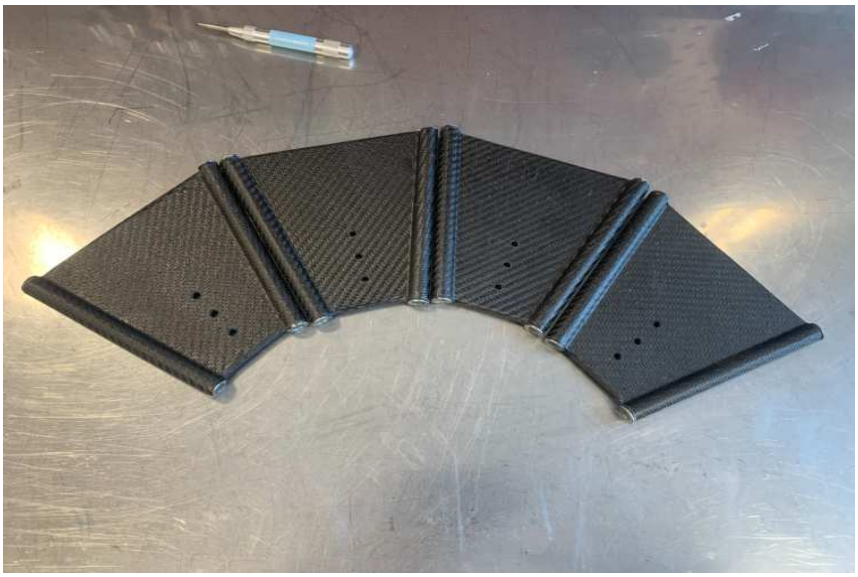


Figure 8: Four lightweight A-arms for the lower portion of the suspension solution made in carbon fibre.

A 5 mm thick aluminium plate serves as the motor mounting interface, providing structural attachment points for the lower wishbones and integrating the suspension with the drivetrain. For ease of construction and manufacturing, the four brackets are screwed onto the motor bracket. 5 mm thick M8 conical washer was used at each uni-ball mounting bracket to allow each wishbone to move freely without interfering with the brackets. Since the lower wishbone is a solid piece, the angle can not be changed and is fixed at 31.2°. While the angle of the upper wishbone can be changed to avoid differing geometries, and adjusting the lengths of the linkages. The angle of the upper wishbone is also kept at 31.2°.

7.3.5 Manufacturing and Assembly

Once the final frame dimensions and the designs of the brackets and the suspension components had been finalised, the frame was manufactured by a local welding shop using aluminium as the selected material. Several suspension components were manufactured through Computer Numerical Control (CNC) machining and were also produced in aluminium. These components include the mounting points between the frame and the shock absorbers, the mounting plate between the control arms and the motors, and the distance plate between the motors and the wheels. An exploded view of the wheel, spacer plate, motor, and mounting plate is presented in Figure 10.

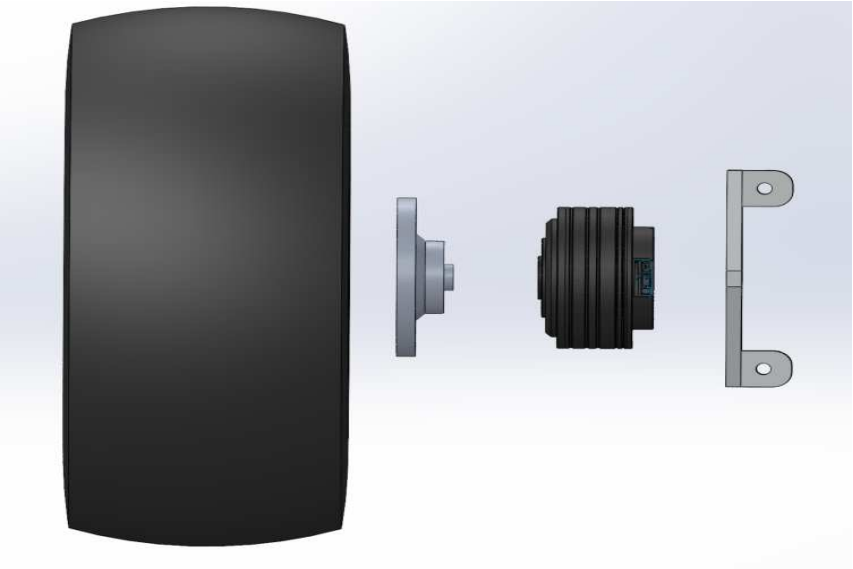
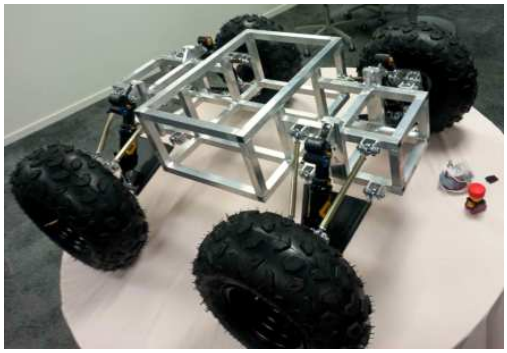
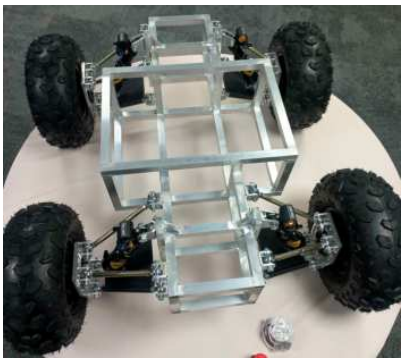


Figure 9: The figure shows (from left to right) a wheel, a distance plate, a motor, and a mounting plate between the motor and the a-arms, with the CNC'd parts being the distance plate (second from the left) and the mounting bracket.

The mounting brackets connecting to the frame and the suspension were fabricated from bent 4 mm aluminium sheets and welded directly to the frame. Furthermore, the final design of the lower control arms was manufactured from carbon fibre. The upper arms consist of separate metal rods that were commercially available, reducing the time required to design and manufacture an upper A-arm. Additionally, the design allows straightforward length adjustment, which simplifies the wheel alignment and enables fine-tuning of camber and toe angles during setup for the wheels. Once all components had been manufactured and the mounting brackets had been welded to the frame, the assembly process was initiated. The suspension was fastened to the frame using M8 bolts and nuts, and M8 fasteners were used between the control arms and the motor mounting plate. Larger M10 fasteners were required to mount the shock absorbers to both the frame and the lower control arms. The motor mounting points utilise M4 bolts; therefore, M4 fasteners were used to secure the motors to the mounting plate and to attach the motors to the brackets connecting the motors to the wheels. In addition to the fasteners, the motor interface includes a central protruding hub that provides primary support for the motor, ensuring that mechanical loads are transferred through the mounting geometry rather than the bolts alone. Finally, the wheels were mounted to the motor brackets using M8 bolts. Figure 10 shows the assembled chassis.



10.1. Final chassis with suspension diagonal view



10.2. Final chassis with suspension top view

Figure 10: The image shows the assembled chassis with the suspension mounted on the frame.

7.3.6 Electrical system

The electrical system provides both power distribution and safety functions for the UGV. The purpose of this section is to describe the system architecture, the included components, and the overall design choices.

All components included in the electrical system are listed in the bill of materials, in the Git repository [105].

The system is powered by a 48 V LiPo battery, which is further explained in Section 7.3.7. A 48 V battery was chosen because the majority of the system's current draw originates from the motors, which operate at 48 V. Power is then distributed to other components such as sensors, the onboard computer, and the router, through a set of DC-DC converters and safety components. The overall system design and setup is illustrated in Figure 28 and further detailed in Appendix E.

Power distribution was implemented using DC-DC converters. These converters provided three different voltage levels within the system. The input voltage of 48 V was regulated to the required output levels, where the output voltage was adjusted during system configuration using a potentiometer. The onboard computer is the Intel NUC 13, which operates at 20 V. All sensors and communication modules interface directly with this unit, which handles data processing and system coordination. The system offers two Controller Area Network (CAN) buses, one for the batteries and the other for the motors. The buses are connected to their own CAN-USB-adapter, which then connects directly to the computer. The thickness of the cables connecting the components in the system together was calculated using a conventional cable thickness calculator. The two most important parameters to be considered were cable length and the amount of current they will carry. This results in the cables closest to the battery requiring a larger cross-sectional area than, for example, the cable to each individual motor.

Safety Features: The LiPo battery requires strict safety measures, as the risk of thermal runaway must be minimised under all circumstances. One potential cause of thermal runaway is regenerative operation of the motors, which happens when the motors are freewheeling or decelerating. The system must account for this condition, as it could pose a risk to the battery. The battery has a charging limit of approximately 50 V, and the system must also be designed to prevent of the voltage exceeding that limit, as doing so could damage the battery. To solve the regenerative power issue, an over-voltage protection circuit was used. This board is an off-the-shelf product that is specifically designed to absorb regenerative energy from DC motors and is called a shunt resistor. It works by monitoring the power line and applying a two-level resistive load during over-voltage conditions. The over-voltage threshold at which the resistive load is applied is set manually, allowing the operator to select predefined values using a rotary switch placed on the board. There is also an auto mode where the shunt resistor automatically sets the threshold 5 V higher than the voltage level on the power line, 3 seconds after system start-up.

Additional protective measures include fuses and diodes to prevent reverse polarity and protect the system against over-current conditions and potential damage. Other operational safety features include an emergency stop button and a main switch for overall system control. The main switch terminates the circuit right after the battery, which means that the system is fully shut down whenever the switch is turned off. The switch is normally used to start and shut down the platform, but can also be used for safety purposes. The battery voltage level can be monitored continuously through the TATTU app or CAN bus, and if the battery voltage reaches unsafe levels, the battery must be disconnected from the remaining circuit, which is done by turning the main switch. The emergency stop is designed to only cut the power to the actuators, according to the machine directive [106]. The stop button cuts the circuit using a contactor. The contactor works as a relay, but is designed to handle higher voltages than a traditional relay. Because the contactor is normally open, voltage must be applied by the control signal for the contactor to close. This means that if any faults appear with the emergency button, such as cable damage, the contactor would open and the motors would stop. This configuration allows for a reliable emergency stop button. When the emergency stop button is activated, the power to the motors is immediately disconnected, bringing the platform to a stop. The chassis is connected to ground, which ensures that currents are safely diverted in case of faults. This improves the overall operational safety by reducing the risk of electrical shock to operators.

The electrical system was designed with modular architecture, making it easy to connect components and to adjust the number and type of components included in the electrical system. Although the components selected for this project are fixed for the current implementation, they may change in the future, as the platform will be used in research settings where flexibility is expected. The majority of the electrical components were assembled on a pair of PCBs. The boards were assembled into one, and this board is referred to as the PDB.

Initially, one PDB was designed for motors and sensors, to centralise power distribution, reduce cable routing complexity, mitigate loose cable connections, and improve electrical system safety in terms of overheating. As the process progressed and component knowledge emerged, the decision was made to design two separate boards. The design choice was mainly made to simplify the soldering process. A thicker

copper layer requires a higher temperature to properly warm up the pads and solder the components. By dividing the boards into two, where one holds power distribution for components that run on higher voltage and currents, the remaining components could be placed on a board with a thinner copper layer. This resulted in one board that handles high voltage components such as the motors and router. The other board manages lower voltage devices, such as the sensors and the computer. The PDB will be explained in detail in the following two paragraphs.

High-voltage PDB: The high-voltage PDB was designed to supply the motors and the router with 48 V. For the motors, the PDB was designed to handle a maximum total current of 120 A (30 A per motor), including current spikes up to 31.9 A according to their data sheet [107]. The router section was designed to handle 20 A, significantly exceeding the router's actual requirements. This oversizing allows the traces and connectors to be repurposed for other high-current applications in future iterations. While the router does require high current, the trace width was selected conservatively based on the available copper thickness to ensure sufficient thermal margin. The motors current values determined the appropriate copper thickness, calculated by an online PCB trace width calculator[108]. Using parameters of 30 A, a temperature rise of 50°C, and an ambient temperature of 25°, the calculations resulted in a copper thickness of $140\,\mu\mathrm{m}$. This thickness allows for narrower traces while maintaining safe operating temperatures. The board was divided into three sections to split the current: one supplying the router and two for the motor system. This was to handle the high currents, the four motors were split between two connectors, which is shown in Figure 11. Connector B1 supplies the left-side motors, while B2 supplies the right-side motors. Both connectors handle current up to 90 A. This division minimises current flow through each connector and reduces the required trace width.

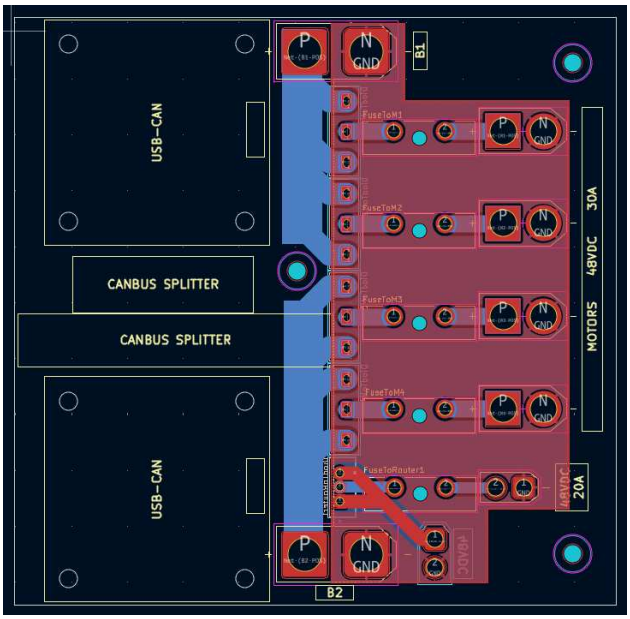


Figure 11: High-Voltage PDB: The power trail, shown in blue, starts at an XT90 connector with an initial trace width of 7 mm, which is then reduced to 4 mm at each Schottky diode, followed by 3mm traces to the fuse holders and further to each motor connector. The router subsystem uses an XT30 connector as input, with a 2.5 mm trace width to the Schottky diode, followed by 2mm traces to the fuse holder and the router connector. Due to lower current requirements of the router, narrower traces are sufficient in this part of the system. The PDB also includes provisions for CAN communication splitters and USB-to-CAN adapters, which are external components installed at a later stage.

Low-voltage PDB: The low-voltage PDB was designed to provide regulated voltages for components that requires less than 48 V. The UGV has components that require 24 V, such as LiDAR, as well as 20 V for the onboard computer and 12 V for other devices. Based on the lower current requirements, a PCB trace width calculator [108] determined that a copper thickness of $70\,\mu\mathrm{m}$ was sufficient to minimise trace width. Voltage regulation is achieved using DC-DC converters with adjustable output voltages controlled by potentiometers. The PDB provides regulated outputs of 24 V for six peripheral outputs, 20 V for two outputs, and a configurable 5 V or 12 V output for three devices. The board consists of three voltage

regulator sections, as shown in Figure 12. Two sections are identical, while the third includes a larger trim function to modulate the output voltage, allowing selection between 5 V and 12 V.

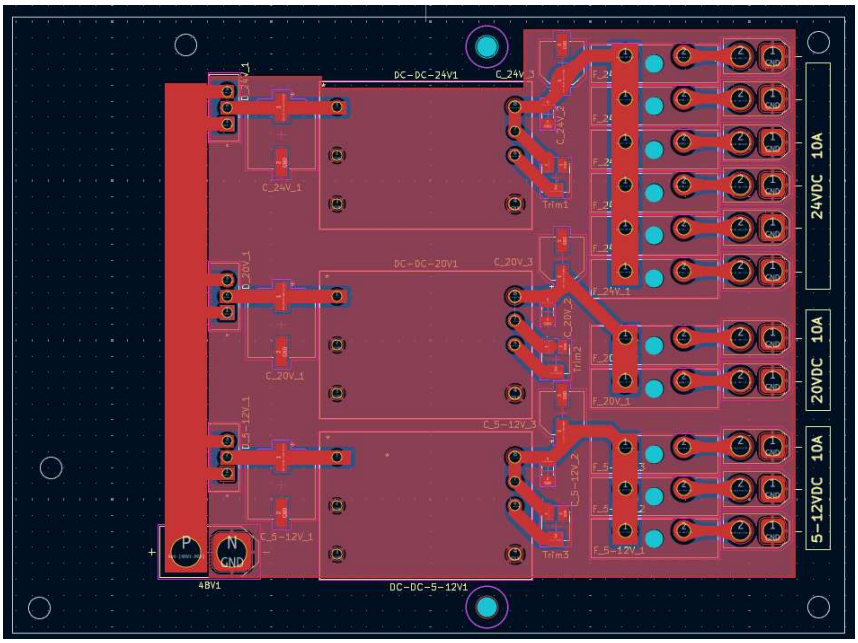


Figure 12: Low-Voltage PDB: The power trail, marked in dark red, starts at an XT60 connector with an initial trace width of 6.6 mm, then 2 mm to each Schottky diode. The trace then passes through an input capacitor to the DC-DC converter [109], followed by two additional capacitors. These capacitors act as a filter for the converter. Below the capacitors, a trim potentiometer is used to set the output voltage. The power trail then continues to the fuse holder and the corresponding output connector. The DC-DC converter is rated for a maximum output current of 10 A, a minimum width of 2 mm is sufficient for this section of the board.

Manufacturing and Testing: After the design was completed, the boards were ordered from two different manufacturers. One manufacturer produced the high-voltage board, and the other produced the low-voltage board, based on their capabilities to handle the required copper thickness, cost, and lead time. The project group decided to order high-quality PCBs for improved board robustness, and to include silkscreen text to indicate routing paths which facilitates troubleshooting.

For testing, a power supply with a maximum voltage of 30 V and a current limit of 10 A was used together with two custom test cables matching the connectors. This was carried out systematically prior to testing the batteries to ensure safety and allow controlled voltage regulation. All DC-DC converters on the low-voltage board were verified and adjusted using a multimeter to ensure the correct output voltages. Additionally, one section was tested with a sensor connected to confirm functionality under load. To validate the functionality of the high-voltage PDB, one section was tested using a motor controlled by a computer, while the remaining outputs were verified through voltage measurements.

7.3.7 Battery

The battery was selected based on the power demand of the four CubeMars AK10-9 motors and the project's operational requirements. More detailed calculation was performed to determine a more appropriate capacity, taking into account the total system mass was 75 kg (including a 25 kg payload), a slope of 30°, drivetrain efficiency of 85 % according to the motor's specification at a torque of 13 Nm [107], and a velocity of 1 m/s. These parameters represent the worst-case scenario defined in Table I, and were used to ensure sufficient endurance over-dimensioning the power source.

The gravity affecting the platform, denoted by F_{grav}, is presented in Equation 1 [110]

$$F_{\text{grav}} = mg \sin\theta = 75 \cdot 9.81 \cdot \sin(30^\circ) = 367.875\ \text{N}, \tag{1}$$

where m is the total mass, g is the gravity constant and θ is the incline. The approximated rolling resistance force, denoted by F_{rr}, was calculated in Equation 2

$$F_{\text{rr}} = C_{\text{rr}}\, mg \cos\theta = 0.1 \cdot 75 \cdot 9.81 \cdot \cos(30^\circ) \approx 63.72\ \text{N}, \tag{2}$$

where C_{rr} represents the rolling resistance coefficient for car tyres in outside field conditions [111]. The total mechanical driving force was calculated in Equation 3 [110].

$$F_{\text{tot}} = F_{\text{grav}} + F_{\text{rr}} \approx 431.59 \text{ N}, \tag{3}$$

The mechanical power required by the platform, denoted by P_{mech}, was calculated in Equation 4

$$P_{\text{mech}} = F_{\text{tot}} \cdot v = 431.59 \cdot 1 = 431.59 \text{ W}, \tag{4}$$

where v is the maximum velocity. The electrical power required was calculated in Equation 5.

$$P_{\text{elec}} = \frac{P_{\text{mech}}}{P_{\text{drivetrain}}} = \frac{431.59}{0.85} \approx 526.33 \text{ W}, \tag{5}$$

where P_{elec} represents the power required for the platform to climb a 30° incline in 1 m/s, and $P_{\text{drivetrain}}$ represents the estimated efficiency of the drivetrain. The required continuous torque per wheel is calculated is in Equation 6

$$T = \left(\frac{F_{\text{tot}}}{4}\right) \cdot r = \left(\frac{431.59}{4}\right) \cdot 0.165 \approx 17.8 \text{ Nm}, \tag{6}$$

where r is the wheel radius for the UGV. The total power consumption of the platform is shown in Table III. Furthermore, the platform integrates various sensors and communication components. While the system is designed to support the integration of sensors, their detailed implementation is outside the scope of this project. However, the associated power consumption must be taken into consideration, as it directly affects the battery selection. R4 in Table I specifies a minimum runtime of 20 minutes; therefore the operating time was set to one hour to provide a safety margin.

TABLE III
TOTAL POWER CONSUMPTION BY THE PLATFORM COMPONENTS.

Component	**Continuous (W)**	**Peak (W)**
LiDAR (2 units)	13	36
Router (4 ports)	120	240
Intel NUC (4 ports)	120	120
Stereo camera	6	7.5
Motors (4 units)	526	6144
Drive module (4 units)	negligible	negligible
Total	**785**	**6548**

	Continuous	**Peak**
Energy (Wh)	785	6548
Battery capacity @ 48V (Ah)	16	137

Equation 7 states the energy calculation, and equation 8 shows how the capacity was calculated. The voltage of the battery was set to 48 V, as that is what the majority of the components require. As shown in Table III, a 48 V, 16 Ah battery is needed to run the platform for one hour.

$$E_{\text{Wh}} = P_{\text{W}} \times t_{\text{h}} \quad \text{where } t_{\text{h}} = 1 \tag{7}$$

$$C_{\text{Ah}} = \frac{E_{\text{Wh}}}{V_{\text{V}}} \quad \text{where } V_{\text{V}} = 48 \tag{8}$$

With these calculations in mind, the chosen battery was a 22 Ah, 48 V battery from Genstattu [112].

Battery box Two fully enclosed aluminium boxes were manufactured to house the batteries in the mid-section of the UGV. The battery boxes are designed to protect the batteries in the event of a crash or other unforeseen circumstances. The battery boxes are also intended to limit exposure to the rest of the UGV in the event of a battery fire. The boxes were manufactured as a two-piece construction, one piece being the housing and the other being the lid. Each piece was fabricated with 2 mm-thick aluminium sheets and is

anodised to be resistant to surface wear. The lower section of the battery box is large enough to house the entire battery and is riveted to keep it from separating. A custom 3D-printed mounting bracket holds the battery connector of the battery to one of the short sides of the battery box. This enables the battery to be connected to the PDB with the battery box fully enclosed. The midsection of the chassis is designed to hold two battery boxes side by side. To simplify the installation or removal of a battery box, the protective carbon fibre panel was made removable. Two raised aluminium rails are fitted to the lower part of the midsection to reduce the friction when installing or removing the battery boxes from the midsection. The rails also act as a cover, keeping the battery boxes from having any direct contact with cables running across the bottom of the midsection

7.3.8 Motors

The Robotnik Summit [61] uses a similar configuration, where the motor is mounted in the wheel, enclosed by the outer tyre. The selected motors for this project: the CubeMars AK10-9 V3.0 KV60 were suitable for this configuration based on several factors, such as weight and decreasing complexity. Using a configuration with one motor mounted in each wheel eliminates the need for a rear axle and the associated weight, and also leads to better traction and power from each wheel. However, it requires a communication link between the motor and the batteries in order to control the power output of each motor and for the battery to supply power appropriately. Although the CubeMars motors support UART and are compatible with the manufacturer's software (CubeMarsTool Upper Computer V3.0), a custom CAN bus communication was implemented due to the limited multi-motor support and insufficient software functionality for simultaneous control of multiple motors. The motors have a rated power consumption of 442 W and a maximum power of 500 W. Additionally, they deliver a continuous torque of 18 Nm and a peak torque of 54 Nm, making them suitable for our application. The motor has a rated no-load speed of 320 Rounds Per Minute (RPM). Furthermore, the motor is lightweight and compact, with a mass of 940 grams, and features dual encoders that enable bidirectional control. It is compatible with the manufacturer's Driver Board V.30 [113], which communicates over CAN bus, allowing seamless integration with the ROS2 framework. The raw data provided by the motor uses Electrical Rounds Per Minute (ERPM), with a range of ± 50000. To convert this to mechanical RPM, the following relation was used

$$n_{\text{mech}} = \frac{n_{\text{elec}}}{p}, \tag{9}$$

where n_{mech} denotes the mechanical shaft speed in revolutions per minute (RPM), n_{elec} denotes the electrical rotational speed expressed in ERPM, and p is the number of motor pole pairs. According to the motor datasheet, $p = 14$.

The motor driver reports velocity as ERPM. This value is converted to mechanical RPM by dividing the number of pole pairs specified in the motor data sheet. The mechanical RPM is subsequently converted to angular velocity and linear wheel velocity for further use in odometry and motion estimation. Although the motor driver accepts velocity commands in a range of up to ± 50000 ERPM, this range represents a generic control limit. Given the motor's maximum mechanical speed of 320 RPM according to the data sheet, and 14 pole pairs, the corresponding maximum electrical speed reaches 4 480 ERPM, even though the driver accepts ± 50000 ERPM.

The CubeMars motor provides a theoretical max vehicle speed of approximately 20 km/h shown in Equations 10 and 11.

$$\omega_{\max} = \frac{2\pi n_{\max}}{60} = \frac{2\pi \cdot 320}{60} \approx 33.5 \text{ rad/s}, \tag{10}$$

$$v_{\max} = \omega_{\max} \cdot r = 33.5 \cdot 0.165 \approx 5.53 \text{ m/s} \approx 19.9 \text{ km/h}, \tag{11}$$

where $n_{\max}$ denotes the maximum motor speed in revolutions per minute (RPM), $\omega_{\max}$ denotes the corresponding maximum angular velocity in radians per second (rad/s), r is the wheel radius, and $v_{\max}$ is the resulting theoretical maximum linear velocity of the vehicle.

7.3.9 System communication

Two separate CAN-buses were designed to ensure stable communication between the batteries, motors, and the computer. The design was tested using a breadboard, jumper cables, and a $120\,\Omega$ termination resistor, verifying the functionality of the circuit. After successful testing, the design was implemented in the system.

The complete system was composed of two ODrive CAN-USB adapter [114], a CAN splitter, and custom-made JST GH cables, together with a 120 Ω termination resistor fitted with a JST GH connector. The CAN-USB adapter was included to convert USB data from the computer into CAN messages and vice versa. CAN splitters were used to connect the different components to the respective CAN-bus. Two instances of this circuit were implemented in the final system, one circuit for the battery communication and one for the motor communication. The communication was chosen to be separate to facilitate troubleshooting and future system extensions. The CAN-USB adapters and CAN splitters were mounted to the PDB. Each motor and batteries are connected to the splitters in a star network using twisted CAN-high and CAN-low JST GH signal cables, with a proper termination applied at the bus endpoints.

The batteries and motors utilise CAN 2.0B with extended frames and a baud rate @ 1 Mbps, whereas the batteries send multi-frame messages, while the motors send single-frame, the structure of the frames can be seen in Figure 13 . The battery transmits eight frames with a frequency of 4 Hertz (0.25 seconds). The

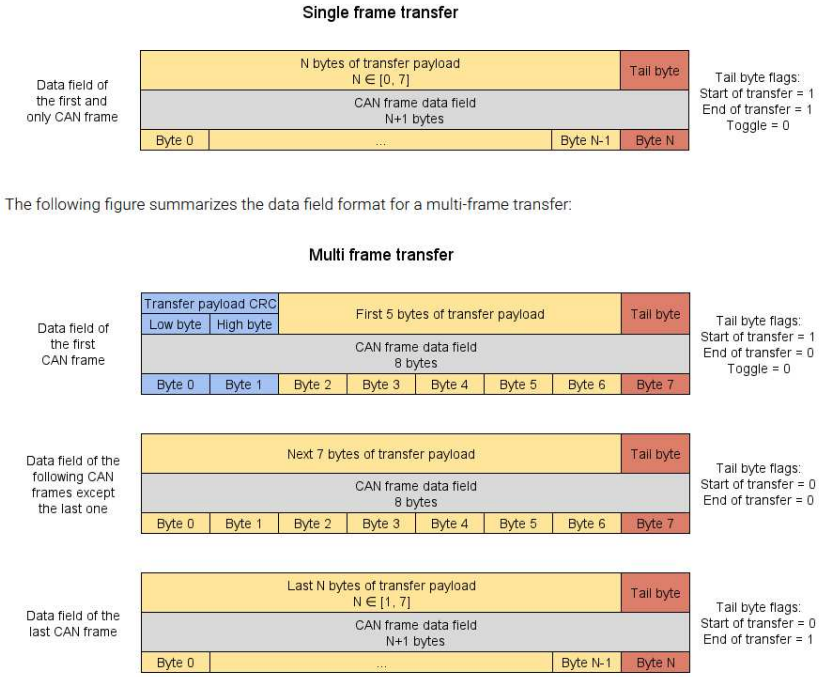


Figure 13: This figure illustrates the composition of a single frame and a multi frame transfer. The motors utilise single-frame transfer, where each frame sent and received contains a content identifier, content, and a tail byte. In contrast, the batteries employs a multi-frame transfer protocol in which the first frame contains transfer payload data, content data and a tail byte. The subsequent frames only contains content data and a tail byte. The built in battery protocol transmit eight frames sequentially, the parameters are always structured in this predefined order. Additionally, this implemented solution, identifiers are unnecessary with the exception of the ordering of the frames being a crucial factor to keep track of.

order of the messages is crucial to decode the information, where the frames lack explicit content identifiers. Therefore, the frame order specifies the content of each frame. The communication protocol and decoder were implemented with this behaviour in mind, ensuring correct message interpretation independent of the number of batteries connected to the bus. The battery script was written to broadcast all parameters from the battery to the computer to ensure safe system operation, without risking damage to either the system or users. The motor script was written using the servo mode of the motors due to the simplicity of implementation and the behaviour of the platform. During system startup, each motor is explicitly set to servo mode before any velocity commands are accepted, ensuring a defined and safe initial state prior to motion. Setting a specific velocity rather than a target angle was decided to be more suitable for this application based on its intended use of teleoperation. The MIT mode was not implemented. The script was written to receive a twist vector from a path-planning or teleoperation node. The script translates the twist vector to motor commands with a differential drive model. Furthermore, the script receives status frames from the motors through the CAN bus, and the information is then transmitted to the computer. This telemetry is used for system monitoring and safety-related checks. This information contains encoder data important to improve navigation and safety-critical information, such as motor temperature.

7.3.10 Sensor mounting design

Restating Table I, the platform shall support modular sensor mounting. This requirement provides flexibility in experimental setups and facilitates modifications and further development of the sensor suite. The mounting interface was designed around three main components: a mounting surface plate, standoff carbon fibre pipes, and fastenings. A mounting plate with multiple, equally spaced through-holes was manufactured and placed on the top of the robot. The holes were distanced to allow fastenings to be mounted with a small footprint without compromising stability. With a fixed spacing of 75 mm along the longitudinal x-axis and 45 mm along the lateral y-axis of the robot body frame, through-holes were distributed across the mounting plate to maximise modularity. The plate was manufactured in 2 mm thick aluminium, with a hard anodised coating, by a supplier. Mounting brackets were designed using SolidWorks, inspired by a GoPro solution [115] and manufactured using 3D printing. The mounting brackets have through-holes that allow the component to be mounted on the surface plate using bolts and nuts. The component also has additional through-holes allowing sensors to be fastened. This design results in a single standardised component, simplifying replacement and re-manufacturing in the event of damage. The last feature of the fastening component is a pipe holder where the 30 mm carbon fibre pipes fit. The solution was completed by attaching carbon fibre pipes between fastenings mounted on the surface plate and the respective sensor, and can be seen in Figure 14 . The length of the carbon fibre pipes was adapted to the requirements of



Figure 14: This image shows the implemented sensor mounting interface. The mounting plate is placed on the central part of the UGV. Custom designed, 3D printed fasteners are fastened to the mounting plate and sensors. 30 mm carbon fibre pipes connects the fasteners between the sensor and mounting plate. Additionally, the image shows a variety of sensors mounted to the platform including a LiDAR, 5G antennas, a stereo camera, a FPV camera and a GPS antenna.

each sensor, and the through-holes of the fastenings were adjusted to fit each specific sensor. Some sensors required adapters to fit the general fastening design and to be mounted at a certain angle; these adapters were 3D printed and designed specifically for these sensors.

7.3.11 Hardware Components

Figure 15 gives an overview of the complete system. The hardware components used in this project are available as a bill of materials in the project's Git repository [105].



Figure 15: This image visualises the primary components composing the UGV platform and their location.

7.4 Testing Phase

The testing phase consisted of evaluating and validating the performance, robustness, and functionality of the UGV's different subsystems. Electrical system testing was emphasised, mechanical tests were conducted, and in addition, teleoperation capabilities were evaluated. Due to the availability of a single platform, testing focus on requirement verification and functional validation rather than limit testing to mitigate destructive risks. The results from this phase provide a basis of assessing the platforms capabilities in terms of the Table I, as well as RQ1 and RQ2 formulated in Section 2.

7.4.1 Electrical System

Individual parts of the electrical system were tested continuously throughout the developing process. This made it possible to verify the functionally of each subsystem. As soon as all pars was delivered and tested the entire system could be tested together, adding one part at a time.

The same method applied for the PDB. To test the boards during soldering of components, sections were completed and validated incrementally. For example, a single DC-DC converter section was soldered and tested for continuity and output voltage before proceeding to the next section including the remaining DC-DCs.

7.4.2 Requirement Validation

The testing of the platform to ensure that all measurable requirements are met was divided into two types of tests. These consisted of physical measurements to assess the platform's physical characteristics and performance tests to evaluate its operational capabilities.

Physical Measurements Physical measurements include R1-R3 in Table I. The weight of the platform was measured by putting the complete UGV on a digital scale. The length, width, and height were measured with a measuring tape at a standstill with the shock absorbers set to 1.72 bar and in normal position. The platform was exposed to increasing payloads, and the reaction of the platform was closely monitored.

Performance Tests Performance tests include R6 - R9 in Table I. The maximum speed of the platform was tested using an application for a mobile phone mounted on the UGV. The phone used its accelerometer to measure the max speed. The incline traversal test was conducted using a ramp with a slippery surface, inclined to 30° shown in Figure 16.1, which the UGV ascended under its own power.



16.1. The platform successfully traversing a 30° incline.



16.2. The platform successfully traversing a 40° lateral incline.

Figure 16: Forward and lateral incline performance evaluation.

Furthermore, R4 and R5 were evaluated by observing the performance of the four-wheel drive system and suspension. Mobility evaluation was conducted in snow-covered terrain, to test the platforms traction, validating for R9. The suspension system was tested by driving over rigid obstacles measuring 10 cm and 17 cm in height respectively, where both obstacles were traversed using a single wheel, as demonstrated in Figure 17. Throughout the lower obstacle traversal, the ground contact of all wheels was observed to evaluate the suspension behavior as illustrated in Figure 17.1.



17.1. The platform maintains full contact on all wheels despite one wheel being elevated.



17.2. The platform traversed a 17 cm toolbox obstacle.

Figure 17: Testing of obstacle traversal.

In addition, the platform tested to climb a plywood surface leaning against a wall, close to a 90° angle with wet tyres as seen in Figure 18. This test was conducted to evaluate the torque-to-weight ratio, and tyre traction on slippery surfaces, as well as the capabilities of the motors.



18.1. Experimental testing of the platform climbing a plywood board leaning against a wall close to a 90 degree angle with wet tyres.



18.2. Traction testing and small obstacle traversal in snow environment.

Figure 18: Various testing procedures.

Finally, a suspension stroke test was performed by comparing the resting length of the shock absorbers, shown in Figure 19.1, with the compressed state shown in 19.2. resulting in a difference of 27 mm, providing an indication of the available wheel travel under load and confirming that the suspension is sufficient for challenging terrain.



19.1. At rest on level ground, the length measured from the top of the shock-absorbers main body to the bottom of the crown, measure to 64 mm



19.2. At rest in a compressed state, with one wheel elevated. The stroke roughly measure to 37 mm

Figure 19: Shock absorber compression testing.

Safety The emergency stop button was tested during electrical system developing using an external stepper motor, to ensure that the functionality worked as intended. Later in the process when the electrical system was assembled, it could be tested using the actual motors integrated on the UGV. The initial tests were performed with the platform elevated from the ground to ensure safe operation. Once the functionality had been verified, the platform was placed on the ground for further testing.

7.5 Documentation

The documentation was created as a user guide for future users, technicians, and administrators [116]. The guide was written as a markdown and hosted on the project's shared Git repository [82] for easy access. The documentation includes more technical information on the components of the UGV, such as the battery safety, a detailed bill of materials [105], the components of the electrical system and how they work. More technical information regarding the motors, CAN communication, and software installation can be found in the documentation [116].

8 Results

This section presents the results obtained from the design, implementation, and testing of the developed UGV platform. The results are structured around the research questions in Section 2. Hardware performance, mechanical and electrical functionality, modularity, and teleoperation capabilities are validated through physical measurements and experimental testing, as explained in Section 7. Together, these results demonstrate to what extent the developed platform satisfies the defined requirements and research questions.

RQ1 aimed to investigate how a UGV-platform could be designed to satisfy the requirements in Table I. The resulting platform fulfils the dimensional requirements, with a width of 86 cm, length of 94 cm, and height of 40 cm, as shown in Table IV. 4WD has been implemented, and all four wheels have individual suspension, which was a requirement from the stakeholders, besides the competition requirements as seen in Figure 20.1 and 20.2. The total weight is 49.2 kg. Additionally, an emergency stop button was implemented as seen in Figure 20.2. As summarised in Table IV, the above section confirms that R1, R3, R4, R5, and R6 were fulfilled.



20.1. A view of the final platform with sensors mounted.



20.2. The final platform with accessible safety features.

Figure 20: The final platform.

TABLE IV
COMPARISON OF REQUIREMENTS AND ACHIEVED RESULTS

ID	Requirement	Result
R1	The platform shall weigh no more than 50 kg.	49.2 kg
R2	The platform should be able to carry a 25 kg payload.	> 25 kg
R3	The total measure (L + W + H) of the platform shall not exceed 250 cm.	94x86x40 cm
R4	The platform shall be a four-wheel drive (4WD).	4WD
R5	The platform shall have suspension of any kind.	Independent suspension
R6	The platform must include an emergency stop button.	Implemented
R7	The platform should be able to keep a max speed of 1 m/s.	3 m/s
R8	The platform shall be able to operate for at least 20 minutes.	> 60 min
R9	The platform must be able to traverse a 30-degree incline.	38.6°
R10	The platform shall be able to move effectively in challenging terrain.	Achieved

Testing confirmed that the platform is able to carry a payload of at least 25 kg without mechanical failure or degradation in mobility. The suspension system remained fully functional throughout testing, with no observable compression. Payload testing was intentionally limited to the minimum required load to avoid potential damage, hence higher payload capacities were not evaluated. Further testing showed a maximum speed of ≈ 3 m/s and incline traversal of $\approx 38^\circ$ surpassing the requirements of 1 m/s and 30°, which are shown in Figure 21 and Figure 22 respectively.

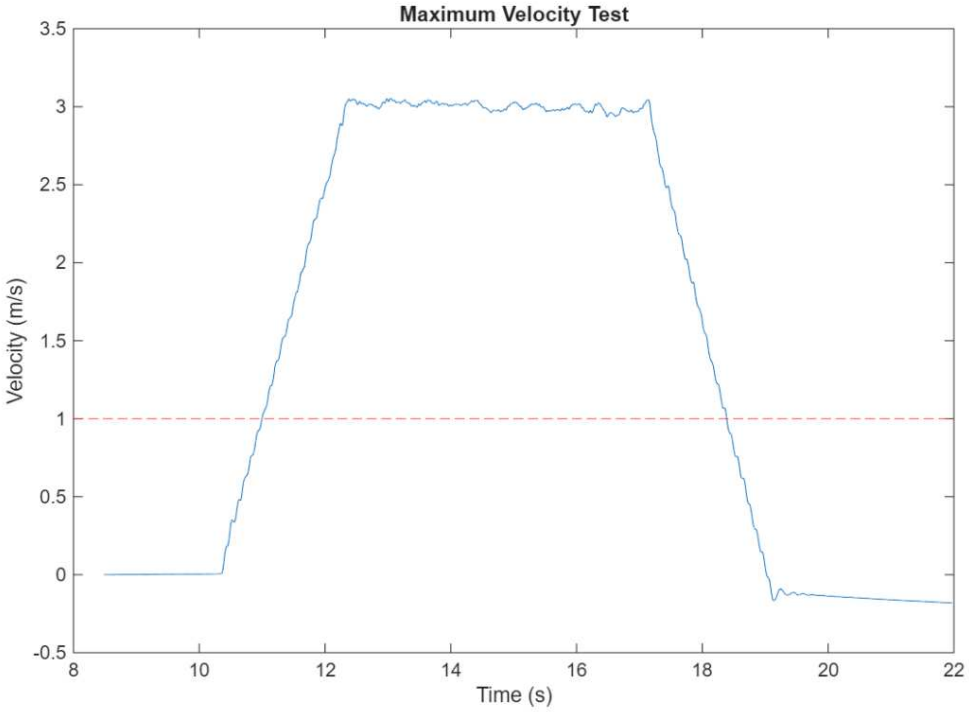


Figure 21: This graph displays the experimental validation of the velocity for the UGV.

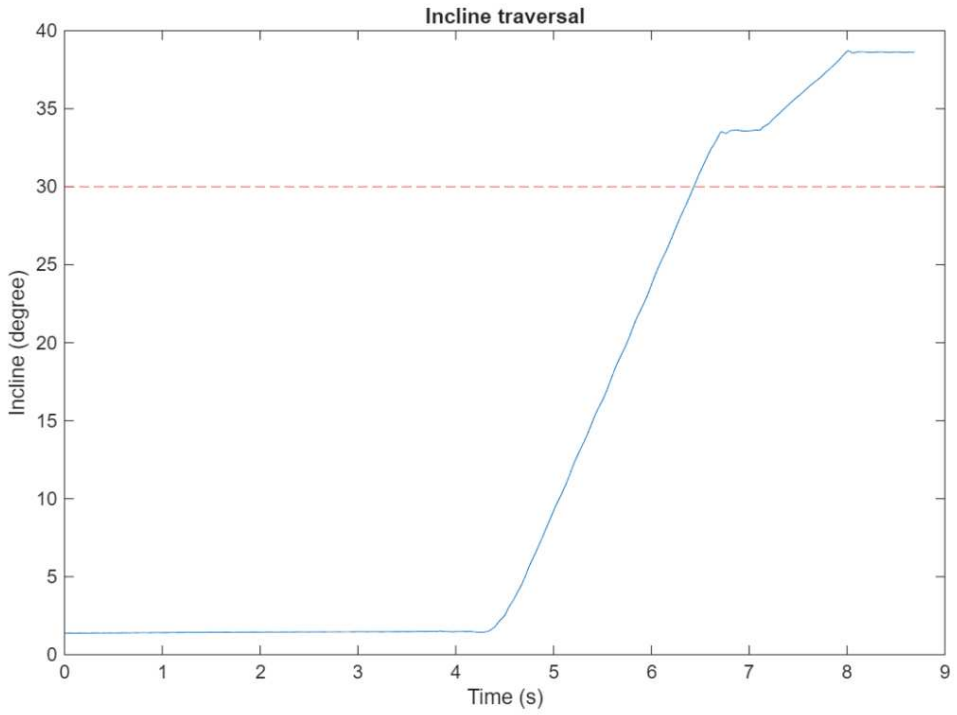


Figure 22: This graph displays the experimental validation of the incline traversal for the UGV.

The platform was tested by driving in snow-covered terrain and ascending a snow-covered incline, demonstrating maintained traction and mobility under low-friction conditions, shown in Figure 23, where it continuously drove for over 60 minutes fulfilling R8.



Figure 23: This figure showcase the capability of the UGV in challenging terrain, specifically in this case in a snowy forest.

Moreover, the platform shows capabilities of overcoming a variety of obstacles utilising the sufficient ground clearance and torque of the motors. Requirements R2, R7, and R9 were fulfilled under the tested conditions conducted in Section 7.4. During testing, the platform successfully teleoperated in outdoor terrain, specifically a forest, while briefly passing behind obstacles, such as trees, provided that communication was maintained fulfilling R10.

RQ2 addressed the development of an adjustable and modular platform with respect to structural configuration and sensor integration. A custom sensor mounting interface was implemented, enabling modular sensor placement and reconfiguration. Structural adjustability is achieved through multiple mounting points for the suspension components, allowing changes to damped positions and suspension geometry, ultimately affecting ground clearance and platform width. Electrical modularity is provided by a modular PDB design, offering multiple regulated voltage outputs and multiple connection points for sensors and peripheral devices, while supporting future sensor integration. The sensor mounting plate, featuring multiple equally spaced through-holes on top of the platform, which facilitates sensor addition, relocation, and removal. The solution is height-adjustable and composed of commercially available or easily manufacturable components.

RQ3 investigated the teleoperated capabilities of the platform, where it was teleoperated using a PS5 controller or a keyboard using scripts running on ROS2 Jazzy. When teleoperated using the PS5 controller, the operational range was measured to 26.2 m, limited by the Bluetooth communication range. The test was conducted outdoors under line-of-sight conditions, and the distance was measured at the point where the connection was lost. When teleoperated via a computer, the operational range is instead limited by the 5G network coverage of the antennas. When communication was lost the UGV stopped accordingly, showcasing safety precautions.

9 Discussion

This section discusses the results of the project in relation to the stated objectives and system requirements listed in Section 2 and Table I respectively. The design choices made throughout the development of the UGV platform were analysed, highlighting the key trade-offs, limitations, and their implications for performance and robustness. Furthermore, the outcomes were compared with existing platforms and alternative design approaches identified and available in the literature, alongside potential improvements.

9.1 Design Strategy and Scope

The initial plan of the project was to buy an existing platform and modify it to satisfy the stated requirements in Table I. This was due to the time limitation of the project, modifying an existing platform avoids time-consuming tasks such as prototyping, manufacturing, and custom component development. During the initial stages, the primary focus was to find suitable platform candidates. This changed as the project's stakeholders desired that a platform be designed and built from scratch, and the direction of the project changed. This change in scope had several important implications for the project. Designing and building a platform from scratch introduces risks as well as possibilities to the project. However, it also resulted in improved learning outcomes by allowing the project group to engage in the full system design process, including mechanical, electrical, and software integration. Furthermore, developing the platform from scratch provided greater design freedom, enabling the creation of a modular solution tailored specifically to the project requirements rather than adapting a pre-existing system. An additional benefit was the elimination of the risk of vendor lock-in to hardware and software that is not open-source.

9.1.1 Project Management

At the beginning of the project, a WBS and a Gantt chart were created based on the assumption that an existing platform would be used and modified. When the project scope changed approximately one month into the project to instead building a platform, the WBS and Gantt were not revised to reflect the new scope. Furthermore, the project team lacked experience in estimating the time required for custom design and manufacturing, particularly for complex tasks such as component sourcing, PDB design and manufacturing, and development of a software library from scratch. These factors contributed to scope creep, as the project gradually expanded beyond its initial objectives without corresponding adjustments to planning and scheduling. Consequently, the schedule presented in Appendix 1.1 suggests that most tasks were delayed, although it no longer reflects the actual scope. Creating a new WBS and Gantt chart based on the new scope, building a platform, would have provided a more realistic timeline and improved resource allocation. This experience highlights the importance of adaptive planning and schedule re-evaluation following significant changes or milestones in a project scope.

9.2 Comparison with Pre-Built Platforms

Comparing the developed platform to the two primary pre-built UGV platforms initially considered during the market research, namely the Robotnik summit [61] and the MAAXII Standard [100], several key differences can be identified. The built platform achieved a lower overall weight while maintaining four-wheel drive and independent suspension, a combination that was not simultaneously fulfilled by the evaluated commercially available alternatives. In addition, the developed platform achieved the greatest ground clearance of 24 cm, while maintaining a mechanically robust structure. The in-wheel motor configuration provides mobility characteristics comparable to those of the Robotnik Summit, with the modular sensor plate adding additional flexibility in sensor placement and future system integration. In addition, the Robotnik summit platform introduces the risk of vendor lock-in and non-open-source software, potentially limiting future development related to autonomy and software implementation, and increasing long-term costs. However, developing the platform from scratch introduced increased development risk and required significantly more engineering effort, compared to modifying a pre-built solution. Commercial platforms benefit from extensive field testing and proven reliability, which the developed platform has not yet achieved. Despite this, the results indicate that building a platform successfully met the project requirements and provides a more adaptable research platform than the pre-built alternatives, while achieving this at approximately half of the financial cost, excluding labour hours.

The UGV developed in this project has a simple and lightweight design, providing significant flexibility in system integration. The middle-chassis allows the addition of another identical battery, thereby approximately doubling the operational runtime. For applications that necessitate longer runtime, the addition of another battery was considered; however, due to the weight limitation, the final design was limited to a single battery for the competition, but two batteries for overall testing. Furthermore, a low weight also allows for future sensors to be installed while still fulfilling R1 in Table I. If necessary, weight can be strategically added to the platform to change the dynamics, for example, to adjust the centre of mass. Despite being built from scratch, the structural design built from aluminium profiles was intentionally simple and avoided complex manufacturing processes, such as milling, contributing to reduced production time and cost.

9.3 Design Trade-offs

The prototype phase was shortened due to the aforementioned changes in project scope and objectives. With additional time and resources for prototyping, multiple design ideas could have been prototyped and evaluated. Such comparisons would have provided valuable insights into the advantages and disadvantages between the alternative solutions, for example, wheeled versus tracked locomotion, both of which are commonly used in industrial UGV applications. Instead, the primary objective of the prototyping phase was to develop a platform that functioned well enough to meet the requirements. A more extensive prototyping phase, including exploring alternative chassis designs, suspension geometries, and overall system configuration, could have resulted in improved performance, robustness, and modularity. Additionally, further testing of the prototypes would not only provide more design options but also better insight into adequate choices of materials and other parameters when manufacturing the final platform. Due to the limited prototype testing, several components and materials had to be chosen with generous margins, potentially increasing cost and weight. The platform consists of a few custom components that facilitate repairs. This will simplify the work for future students and potential researchers when using this platform in various experiments and projects.

9.3.1 Suspension

A double a-arm suspension system was chosen, firstly for the relatively simple design with regard to the geometry, and because this suspension system does not change the wheel angle during suspension travel. Furthermore, if a MacPherson suspension were to be chosen, with the supplied shock-absorbers provided by Öhlins. Even more custom modifications would have been needed to make the shock-absorbers a structural component in the suspension geometry, rather than a load-bearing component.

Testing the suspension system revealed that the initial shock absorber configuration was too stiff relative to the platform's mass and intended operation. This highlighted the importance of proper suspension tuning in lightweight UGV platforms. Without the possibility to adjust the shock absorber characteristics, a different model or different suspension geometry would have been required. For example, a push-rod solution configuration was therefore briefly considered as a potential mitigation strategy. Following that, Öhlins modified the shock absorbers to better suit the platform configuration. The modification involved drilling a small hole in the pneumatic shock absorber to alter its compression characteristics, thereby reducing stiffness, and subsequently adding a conventional spring to compensate for the reduced rebound force. This changed the characteristics and thereby rendered the shock absorbers well-suited for our platform configuration.

Furthermore, each wheel is equipped with suspension that reduces vibrations in the chassis and allows for greater obstacle traversing. Reducing vibrations is a key feature to ensure that the sensors provide qualitative data, which in turn improves the performance of the UGV's navigation and control algorithms. Additionally, reduced vibrations lowers the risk of imposing mechanical issues such as loosening fasteners and bolts, disconnected cables, and overall stress to the structure.

9.3.2 Modular Sensor Plate and Mounts

The objective of designing a modular sensor mounting solution was achieved; however, the implementation involves several trade-offs. Since the sensor mounting plate contains multiple through-holes, additional measures were required to prevent water or debris from entering the chassis. This was addressed by mounting a sealed intermediate plate directly to the chassis and positioning the perforated sensor plate above it using spacers.

The sensor mounts functioned as intended during testing, but their fabrication using 3D-printed plastic introduces a weakness regarding long-term durability. To reduce the risk of mechanical failure, these components could be manufactured in metal using CNC machining in future iterations, much like many of the existing components were manufactured. In its current configuration, the modularity of the sensor system primarily allows repositioning of the existing sensors across the mounting surface. The integration of additional or different sensors would likely require new pass-throughs for cable, risking the requirement of drilling additional holes in the chassis to route sensor wiring to the electrical system. As such, while the mounting solution provides modularity, full plug-and-play scalability is not possible with the current cable routing design and size of the front and rear ends of the chassis.

9.3.3 In-hub motors

The platform employs one motor mounted in each wheel, enabling individual control. This configuration is not only effective when navigating rough environments, but it also leaves more room for sensors in the chassis since the motors are mounted in the rims of the wheel.

9.3.4 Ground Clearance

The UGV provides sufficient ground clearance of 24 cm, which will be utilised when navigating in rough terrain. The ability to adjust the ground clearance increases the UGV adaptability to different types of environments, reducing the risk of becoming obstructed by terrain features such as dense vegetation, uneven ground, or similar obstacles found in such terrain.

9.4 PDB Design Limitations and Improvements

The current design of the PDB protects the batteries from the power generated by the motors during braking or rolling, where power is dissipated as heat in the current solution. The PDB functionality could be further developed to use the generated power to charge the batteries, thereby increasing the efficiency of the platform by extending the runtime. Furthermore, utilising the generated power rather than converting the energy to heat would also reduce the system temperature.

Due to limited experience in circuit board design, only one version of PDBs was manufactured. This proved problematic when design flaws were discovered during assembly and testing. The first issue was identified during a team discussion with a supervisor after all components were hand-soldered to the high-voltage PDB. The discovery was that the Schottky diodes mounted on the PDB to ensure correct current direction and protect the battery were not fulfilling their purpose. The problem was that if the motors generated current during regenerative braking, there was no return path for the current, and the over-voltage protection could not prevent potential damage. Luckily, the issue was discovered before full installation, so the diodes could be removed and a bridge added to complete the circuit. A second design flaw was discovered during router testing. The input connector for the router on the high-voltage PDB had its ground pin connected to the power trace instead of ground. This required switching the cable connections, which is poor practice and increases the risk of connection errors.

9.5 Testing and Validation Limitations

A single platform was built in this study, which limited testing in certain cases, such as maximum payload and extended runtime evaluation. As only one platform was available, destructive and high-risk tests were deliberately avoided to prevent damage to the system. The platform was tested to verify that the requirements stated in Table I were fulfilled with a safety margin. For example, the runtime requirement (R8) was exceeded, as the platform operated continuously for an hour while the battery state of charge remained within the recommended range of 20-80%. Testing beyond this range, such as discharging the battery from 100% to 0% was not conducted, as this is detrimental to long-term battery health. A similar conservative approach was applied to all degradation-related tests.

9.6 Chassis Design Limitations

Due to the development process discussed at the beginning of this section, certain designs and platform components could have been more thoroughly considered. The size of the chassis provides sufficient space

to mount all current necessities, but it limits flexibility for mounting additional hardware. One major limitation is the available space for the PDB and associated cables connecting all systems. Additional space would have simplified cable management and facilitated an easier troubleshooting of the electrical system if necessary. Cable management is crucial to troubleshoot the electrical system and to minimise signal interference for communication, such as CAN, and to reduce the risk of damaging cables. The decision to implement two batteries was made after a substantial number of design decisions had been taken. A feature to include a second battery was added to increase the total runtime and facilitate various future research beyond the ARES Makeathon. However, this had a negative impact on the possible mounting area of other components and the possibility of adding future components with a large footprint. If the chassis design were to be revised, the central part of the body would be broadened to improve cable management and allow for future space-consuming components. With the current design, one possible mitigation would be to remove one of the batteries. While it would marginally increase the process of battery replacement, the platform currently lacks hot-swapping capability, meaning the dual-battery configuration provides limited operational advantage.

The platform has a symmetric design, and the use of bidirectional motors allows for equal performance regardless of direction in terms of forward-driving and reversing. However, this symmetry is limited to the load and the weight of the components mounted to the chassis. The front and rear sections of the chassis house the PDB on one side and the computer and router on the opposite side. The weight difference on these components is close to negligible from a performance perspective. Furthermore, the batteries are mounted in the centre of the chassis add approximately 6 kg each to the platform, which moves the centre of mass of the platform closer to the middle; thus making the platform more stable. However, if one battery were to be removed, the centre of mass would be shifted to either the left or right side, introducing instability by shifting the centre of mass. This could be solved by installing multiple mounting possibilities for the battery compartments, for example, having brackets to lock a battery in the centre of the flooring. Other types of weight distribution techniques should be considered to increase the platform's adaptability to various environments and tasks.

9.7 Teleoperation using ROS2

RQ3 investigated how the developed UGV platform could be teleoperated using the ROS2 Jazzy framework in challenging terrain. The results demonstrate that the platform can be reliably teleoperated using the implemented ROS2-based control interfaces and a handheld controller, enabling real-time manual control of the platform during field testing. The teleoperation tests were conducted using a PS5 controller, where the operational range was limited by the Bluetooth connection, where a maximum range of 26.2 m was measured under best-case outdoor conditions. The teleoperation capability remained functional during non-line-of-sight conditions, although no quantitative evaluation was performed under these conditions. Although computer-based teleoperation was implemented by a parallel project group as part of the system architecture, it was not empirically evaluated in this project. Consequently, the results are limited to short-range teleoperation using a handheld controller.

10 Conclusion

The purpose of this project was to build a UGV platform with a number of physical and performance requirements. Furthermore, a modular and adjustable sensor mounting interface and teleoperation have been implemented as a solution to answer RQ 2 and 3, respectively.

The platform was designed to compete in the ARES Makeathon and to be used in various research areas, such as autonomous navigation. The practical work to deliver a platform was divided into four phases: design, prototype, manufacture, and test. The chassis and suspension solution was designed using SolidWorks [78], while KiCad [80] was used to design the PDB. The design was tested using temporary 3D printed parts and aluminium profiles, aluminium profiles for the frame, and 3D printed parts for the suspension and PDB. Once the prototype phase reached a satisfactory level, the manufacturing phase started.

The resulting platform was robust, mainly composed of components in aluminium and carbon fibre. It has four wheel-mounted motors, each with individual adjustable suspension. "Off-the-shelf" components were preferred over custom components. Easily replaceable components and a simple symmetric design of the chassis facilitate reproduction and repairs. A custom sensor fastening plate was implemented to enable simple addition, relocation, and removal of various sensors. Carbon fibre tubes were used to adapt the height of each sensor, and 3D printed fastenings were used to mount the pipe to the sensor and the plate. Carbon fibre tubes were chosen since they are both lightweight and durable. Finally, teleoperation was implemented. The platform can be controlled using a wireless PS5 controller or a computer were the latter has a range limited to the 5G connection of the platform. The PS5 controller has approximately a range of 26 m. The UGV platform fulfils all key requirements listed in Table I, achieving a total mass of 49.2 kg, an operational runtime exceeding 60 minutes, a maximum velocity of 3 m/s, and the ability to traverse inclines steeper than 30 degrees.

11 Future Work

To further develop the platform to enable more complex tasks, an arm could be implemented. The addition of an arm makes the platform more versatile and facilitates various tasks such as gathering samples, deploying, and the remote movement of objects. Previous tasks in the ARES Makeathon required an arm to perform key objectives such as retrieving water samples. Even though the next set of tasks for the 2027 competition has not been published as of the date of this report, an assessment that a task with a manipulator being present is not improbable.

Improvements to the battery solution could be to design connectors built into the battery compartment, inserting or removing the battery box connects and disconnects the batteries, which simplifies the handling of the batteries and increases safety. If the battery rises in temperature, the battery could swiftly be removed to protect different subsystems, but also the UGV as a whole. Additionally, implementing a hot-swap solution would further extend the runtime with eliminated physical interaction to switch the active battery. Thus, maximising the efficiency of the current two-battery solution.

With the weight restriction of 50 kg, a rework of the suspension location could be beneficial to increase the width of the front and rear chassis sections without an increase in overall width of the UGV. This can be done by relocating each shock absorber and utilising a push rod solution connecting the shock absorbers to each motor bracket. With the use of a push rod solution and relocation of each shock absorber, a lever arm can be used between the push rod and the shock absorber to increase the motion ratio. This will let the shock absorber operate more efficiently, with less force needed to compress the shock absorber.

Vehicle status for the user could be implemented through LED strips that visually indicate the different driving modes of the UGV. For example, using green for autonomous mode and yellow for teleoperation mode would allow the operator to quickly distinguish the current state of the system. Additionally, assigning a distinct colour to error mode - such as red - in reference to the error flags provided by the motors and batteries. This would enable the user to immediately identify faults.

An extension of this UGV is the development of an actively stabilised sensor platform to stabilise the sensors for more reliable data. In the current system, sensors are rigidly mounted to the chassis, making them susceptible to vibrations caused by the nature of uneven terrain. This can result in unstable point clouds and sensor reliability, degrading the overall quality of the sensors. Future work could focus on designing and integrating a two-axis gimbal-based stabilisation mechanism to actively compensate for the chassis roll and pitch by employing a dual Inertial Measurement Unit architecture; one mounted on the vehicle chassis, and another on the sensor platform. Having two IMUs enables relative orientation with sensor fusion techniques and could be used as a feedback controller to maintain a near-horizontal sensor orientation regardless of the UGV's tilt. Currently, the safety feature capabilities consist of a dead-man switch on the handheld controller, a physical emergency stop button and main switch, and error flags implemented in the software. As a potential improvement, a remotely controlled power switch for the motors could be implemented to enable safe and immediate shutdown in case of emergency. Furthermore, the development of a dedicated display interface presenting system status and error flags, together with improved visualisation of telemetry data, would enhance the software and overall operational awareness.

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A Project Management Documentation

1.1 Gantt Chart

The Gantt chart for the project is shown in Figure 24.

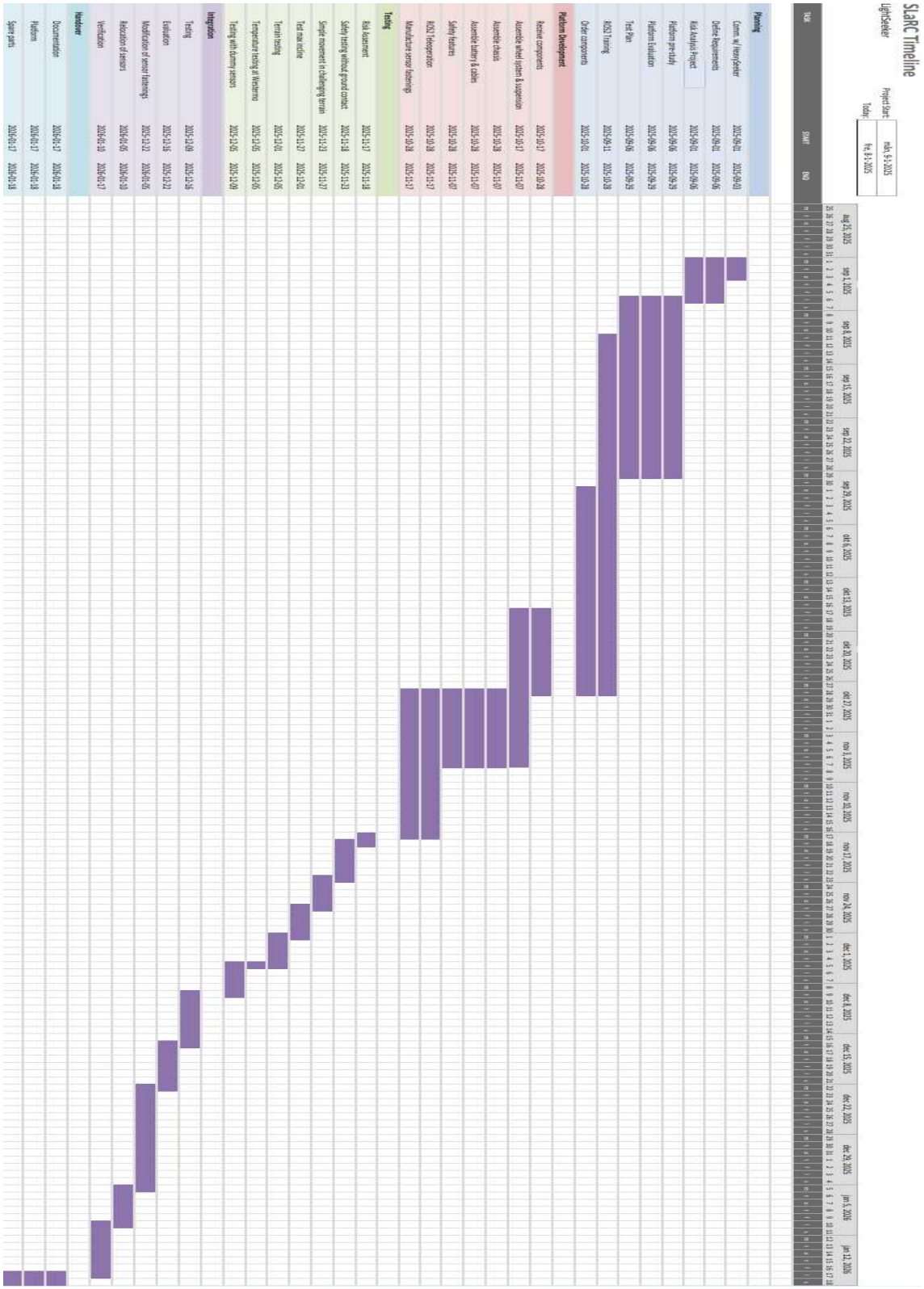


Figure 24: Gantt Chart

1.2 Work Breakdown Structure (WBS)

The work breakdown structure for the project is shown in Figure 25.

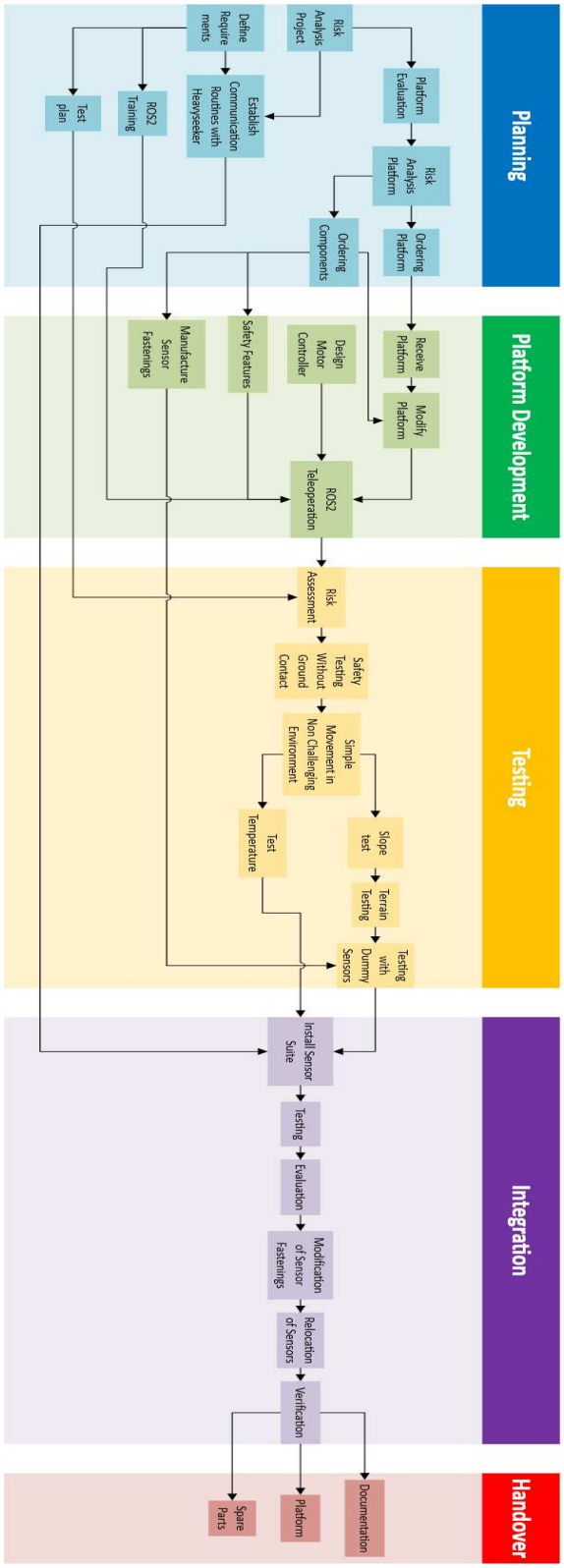


Figure 25: Work Breakdown Structure

B Planning and Risk Analyses

2.1 Project Risk Analysis

The risk analysis of the project is shown in Figure C.

C Risk Assessment

Risk Analysis for the Project

Consequences Rating Guidelines:
1 - No/negligible consequences.
2 - Minor or no time/cost delays.
3 - Moderate time delays and/or cost concerns.
4 - Significant time and/or cost delays, with minor to moderate quality concerns.
5 - Project risks failure and/or significant quliaty concerns.

Score	Risk Level
0 -> 8	Low
9 -> 18	Medium
19 -> 25	High

ID	Title	Description	Date	Probability	Consequences	Risk level	Owner	Mitigation measure	Deadline
Risk 01	Project delays	The project may fail to meet planned milestones due to delays in deliveries, integration, or testing.	2025-09-01	3	4	12	LightSeeker	Introduce regular checkpoints, prioritize critical tasks, reallocate resources when needed, and adhere to the GANTT schedule as closely as possible.	2026-01-18
Risk 02	Insufficient communication between groups	Misunderstandings regarding responsibilities, interfaces, or timelines could cause inefficiency and integration issues.	2025-09-01	2	3	6	LightSeeker & HeavySeeker	Ensure continuous communication with HeavySeeker by scheduling regular meetings and documenting shared requirements to avoid misunderstandings.	2026-01-18
Risk 03	Technical limitations of the platform	The platform may not meet all competition requirements (e.g., 30° incline, runtime, maximum size/weight).	2025-09-01	2	4	8	LightSeeker	Conduct early tests of critical requirements and maintain design margins (weight, dimensions, power).	2026-01-18
Risk 04	Resource shortage (time & personnel)	Illness and high workload can lead to delays.	2025-09-01	2	3	6	LightSeeker	Build time buffers into the schedule and cross-train team members.	2026-01-18
Risk 05	Resouce shortage (material &manufacturing)	Delivery problems and machine unavailability can lead to delays.	2025-09-01	3	4	12	LightSeeker	Plan material ahead and keep backup work for delays.	2026-01-18
Risk 06	Safety hazards during testing	There is a risk of damage or injury during testing (collision, flipping, overheating batteries).	2025-09-01	2	5	10	LightSeeker	Define safe test zones, use an emergency stop system, and perform the testing in a controlled manner. Do not perform testing under rainy conditions.	2026-01-18
Risk 07	Documentation and reporting not completed on time	The group may fail to deliver complete documentation and the final report by the deadline.	2025-09-01	2	5	10	LightSeeker	Begin writing early, assign chapter responsibilities, review progress regularly, and revise drafts toward the end of the project.	2026-01-18
Risk 08	Software integration issues	ROS versions, libraries, or drivers may not be fully compatible with the platform.	2025-09-01	2	3	6	LightSeeker & HeavySeeker	Test software integration early, use TurtleBots if needed, and maintain version control.	2026-01-18
Risk 09	Hardware failure under load	Motors, sensors, or controllers may fail under load after weight reduction modifications.	2025-09-01	2	5	10	LightSeeker	Keep spare components available and conduct stress testing.	2026-01-18
Risk 10	Power limitations	Battery runtime may be shorter than expected.	2025-09-01	2	4	8	LightSeeker	Monitor consumption during tests and prepare backup batteries.	2026-01-18
Risk 11	Conflicts within LightSeeker	Different priorities or misunderstandings may slow progress.	2025-09-01	1	4	4	LightSeeker	Prioritize conflict resolutions, and frequent status check-ins.	2026-01-18
Risk 12	Budget overruns	Costs of additional components or repairs may exceed budget.	2025-09-01	1	4	4	LightSeeker	Prioritize essential purchases and have alternatives.	2026-01-18
Risk 13	Supplier dependency	Delays or lack of documentation from external supplier may slow integration.	2025-09-01	2	3	6	LightSeeker	Secure agreements, request documentation and plan alternatives.	2026-01-18

Figure 26: Project Risk Assessment

3.1 Platform Comparison

The risk analysis of the project is shown in Figure D.

D Platform Comparison

	Robotnik Summit		Robodyne Maxxii Standard
Leadtime	2-3 Weeks		4-5 Weeks
Clearance	~13 cm		~18 cm
Max speed	3 m/s		1 m/s
Weight	65 kg		60 kg
Runtime	5 h		4 h
Locomotion	Wheels		Tracks
Suspension	Yes		No
Cost	~20 000 EUR		~13 500 EUR
	Motivation/Comparison		
Leadtime	Robotinik Summit has a shorter leadtime and is in stock. Robodyne does not have one in stock and can delay an already longer lead time.		
Clearance	Both have high clearande relatively speaking, although too high clearance might result in the platform tipping during steep inclines. Both have low center of mass.		
Max speed	Robotnik's platform is capable of going 3 m/s and will be able to be restricted. Robodyne's platform can achieve a max speed of approximately 1 m/s, and itmight not be able to reach this in all conditions, assuming this max speed is without payload.		
Weight	Both are expected to reach the 50 kg weight limit. Robotnik has had projects with weight reduction; awaiting customer references. Both platforms will be somewhat equally hard.		
Runtime	The aim is 1 - 1h30m runtime, Robotnik's 5 hours and Robodyne's 4 hours leaves little to be desired. Both platforms can have weight reduction in regard to battery.		
Locomotion	Tracks are good for heavy platforms to spread the weight, although since our platform will weigh less than 50 kg these benefitsare less applicable. Furthermore, tracks are heavy and wheels are easier to repair/replace.		
Suspension	Suspension is prefered to reduce vibrations. Sensors such as cameras, and LiDAR are susceptible and perform worse to vibrations, making this a vital. Robotnik's platform suspension solution is both efficient in weight and by capacity while Robodyne's platform does not have a suspension at all.		
Cost	Robotnik is more expensive, but will give better leverage and probability to complete the project within the given time-frame.		
General	For Robotnik we are looking in to if it is possible to purchase the wheels/motors and suspension and the make a custom frame to attach them to in order to reduce cost.		

Figure 27: Platform Comparison

E Electrical System Setup

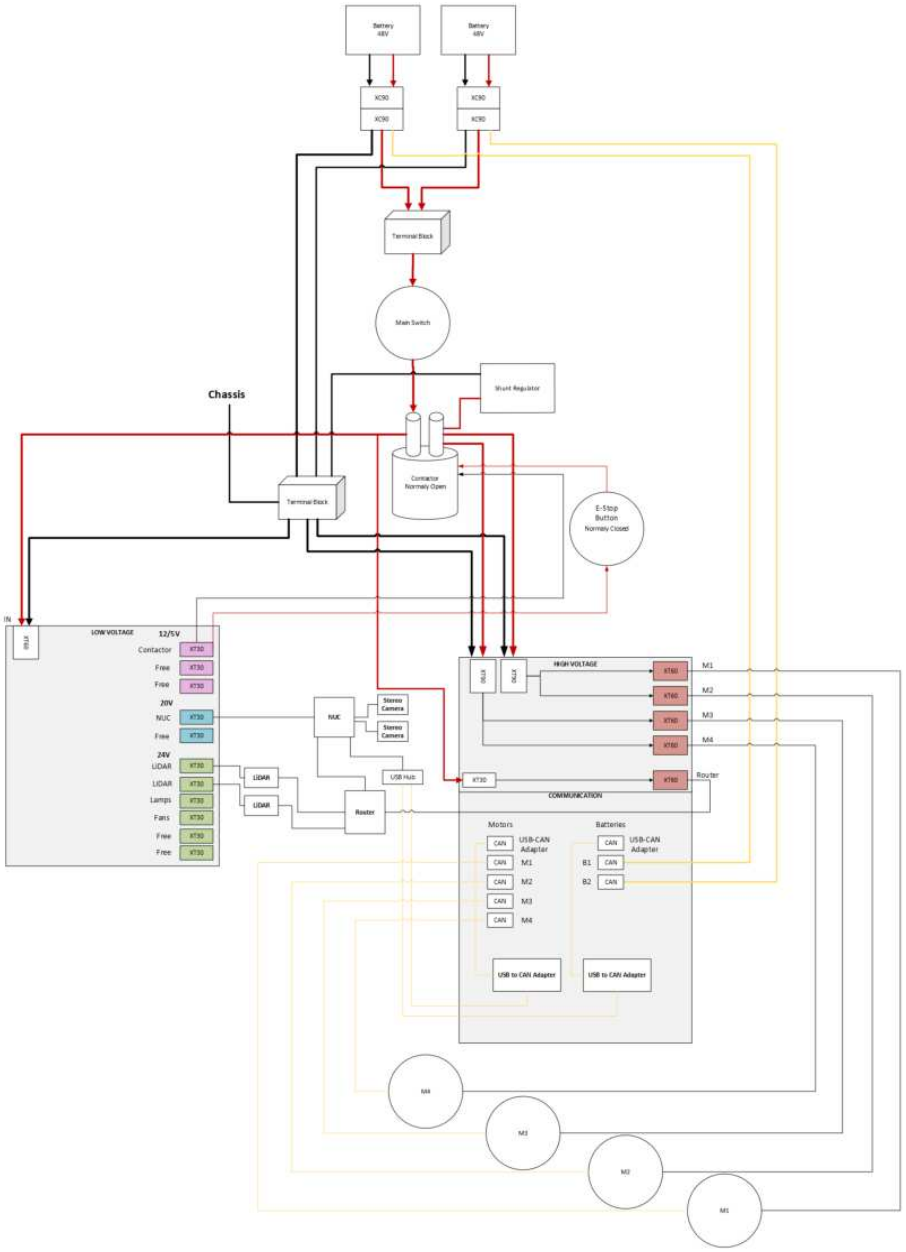


Figure 28: This is a complete illustration of how the electrical system is designed and connected.